

2

Human Hearing and Ultrasound Scanning

In this chapter, we will learn about

- the structure of the human ear and the hearing process
- the properties of ultrasound
- ultrasound scanning

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2.1 Human hearing

Our ears are the sensory organs responsible for hearing. The ear contains the smallest bones in our body, each bone being just a few millimetres long. Although small in size, they are indispensable for hearing sound. Do you know why?

Refer to p.60–62.



Fig 2.1a How can we hear sound?

Flipped classroom
Watch a video and answer the questions.

→ This simulation shows a 3D model of the ear. It is available on OUP web in PC version (.exe) only.

Simulation

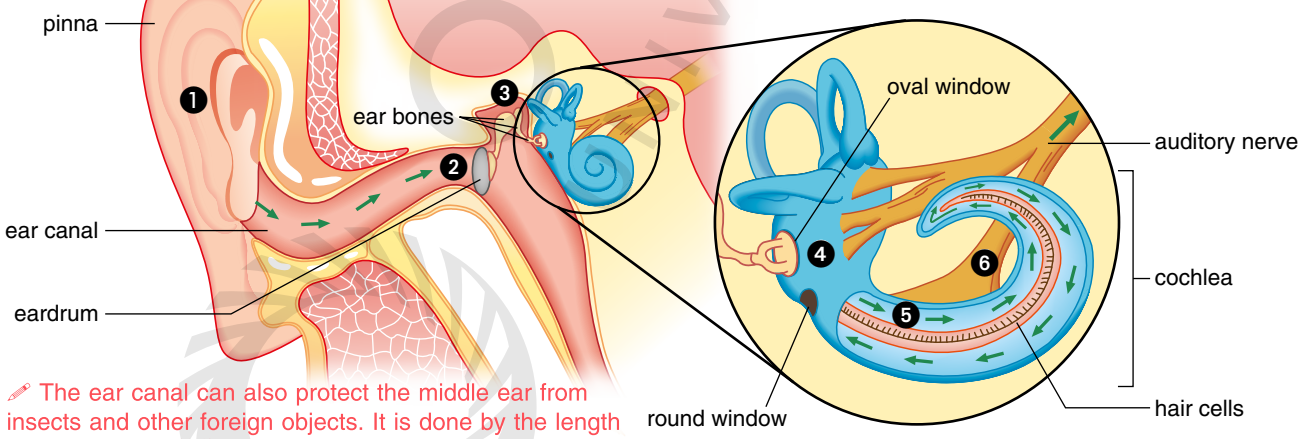
1 Mechanism of hearing

In junior science, we learned about the structure of our ear and how we hear things. Let us have a review.

Pre-study **The human ear**

The figure below shows the basic parts of the human ear and gives a brief description of the hearing process.

Point out to Ss that this figure is not drawn to scale. The inner ear and the middle ear are significantly smaller than the outer ear in real life. The ear bones and the cochlea are smaller than 1 cm each. It may be interesting for Ss to note that the sizes of ear bones are generally fixed at birth. They are the smallest bones in the human body.



The ear canal can also protect the middle ear from insects and other foreign objects. It is done by the length and tortuosity (彎曲) of the narrow ear canal as well as the hair and production of wax.

① The pinna collects incoming sound waves and tunnels them through the ear canal towards the eardrum.	② The sound waves exert a changing air pressure on the eardrum and vibrate it.	③ The ear bones (or ossicles) connected to the eardrum are consequently set in vibration. The vibration is in turn passed to the oval window.
④ The oval window vibrates the fluid in the cochlea.	⑤ The hair cells in the cochlea detect and convert the vibrations into electrical signals.	⑥ The auditory nerve carries the signal to the brain for interpretation.

In the following section, we will look more closely at how the **middle ear** and the **cochlea** facilitate hearing.

DSE goal

Describe pressure amplification in middle ear.

DSE exam

13(2)Q4.4, 15(2)Q.4(a)

✎ Ts may point out congenital malformation (先天畸形) of the outer ear leads to congenital conductive hearing loss. It occurs in 1 in 10 000 births.

a Pressure amplification in the middle ear

The **outer ear** is air-filled while the **inner ear** is fluid-filled. If sound waves directly transmit from the outer ear to the inner ear, only about 0.1% of the sound energy can be transferred through while most of the energy will be reflected back. This is similar to the case if you are swimming and someone yells at you from above the water (Fig 2.1b). In such case, you can hardly hear that person.

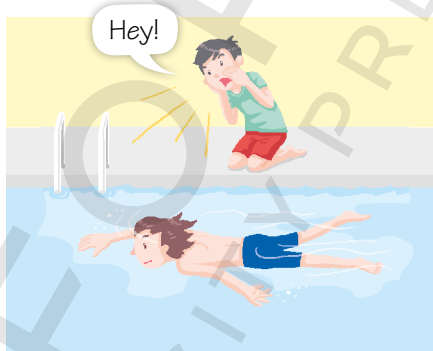
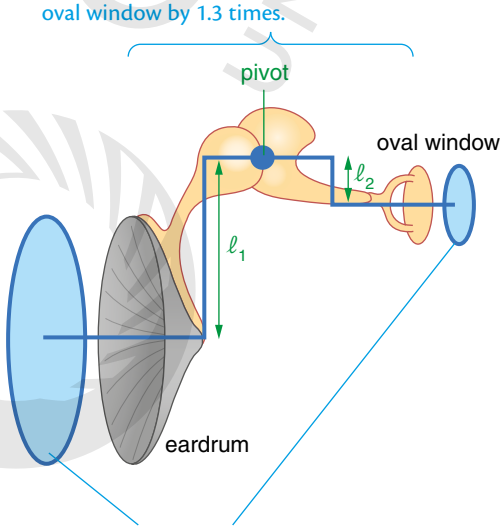


Fig 2.1b *Sound is hardly transmitted from air to water.*

The middle ear plays an important role in solving this problem. It greatly reduces the drop in sound intensity by two pressure amplification mechanisms (Fig 2.1c).

Lever action of ear bones

The three ear bones are interconnected to work as a lever. Together, they amplify the force from the eardrum to the oval window by 1.3 times.



photos of three ear bones



Area ratio of eardrum to oval window

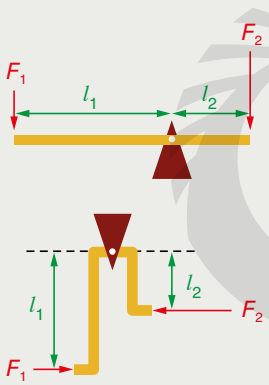
The effective area of the eardrum is 60 mm² while that of the oval window is 3 mm². The area on which force is exerted is decreased by 20 times.

Fig 2.1c *Pressure amplification in the middle ear.*

Know more

Lever systems

Lever systems may take many different shapes. No matter how the shape of the system looks like, it is important to note that the net moment of force is equal to zero when the system is at equilibrium.



For the above two cases in equilibrium,

$$F_1 l_1 = F_2 l_2$$

middle ear 中耳 cochlea 耳蝸 outer ear 外耳 inner ear 內耳

Sound intensity and sound pressure are related as follows:

$I \propto p^2$

The sound intensity level L can be expressed as:

$$L = 10 \log \left(\frac{I}{I_0}\right) = 10 \log \left(\frac{p^2}{p_0^2}\right)$$
$$= 20 \log \left(\frac{p}{p_0}\right)$$

Therefore, an increase in 26 times in sound pressure corresponds to a (20 log 26 =) 28 dB rise in sound intensity level.

Recall that pressure is defined as the force applied per unit area. Since the force is increased by 1.3 times and the area is decreased by 20 times, there is an increase in pressure by a factor of 1.3 × 20 = 26 (Fig 2.1d).



Fig 2.1d *Combined effect of the lever action and the area ratio of the eardrum to the oval window.*

Pressure amplification in the middle ear is a combined effect of

- **the lever action of the three ear bones (or ossicles) and**
- **the area ratio of the eardrum to the oval window (with the area of the eardrum much larger than that of the oval window).**

Google Forms Microsoft Forms

Quick check 1

→ E-version in OUP Exercise Platform is also available.

1 Complete the labels for the basic structures of an ear (Fig a).

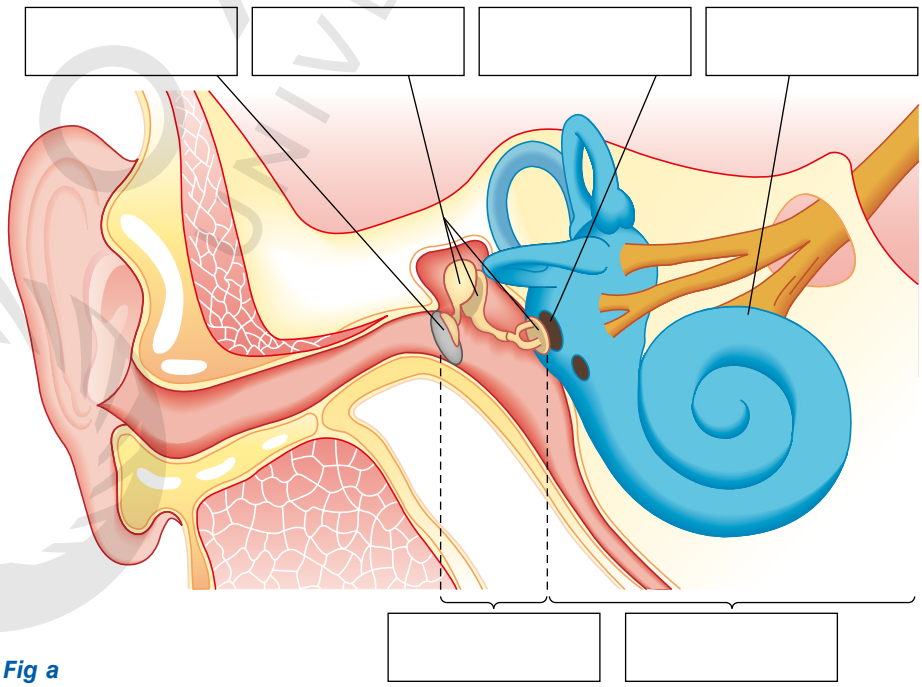


Fig a

2 The pressure of vibration transmitted to the inner ear is amplified as the area of the oval window is (smaller than / the same as / larger than) the area of the eardrum.

ear bone / ossicle 聽小骨　eardrum 鼓膜　oval window 卵圓窗

DSE goal

Realize response of inner ear to incoming sound waves.

DSE exam

16(2)Q4.3, 17(2)Q4.2

✎ The inner ear consists of a membranous system embedded in a bony outer shell.

b Converting mechanical vibrations into electrical signals

The cochlea in the inner ear has an upper channel and a lower channel, in which sound waves travel through. The **hair cells** that detect the sound waves are located in the middle channel (Fig 2.1e).

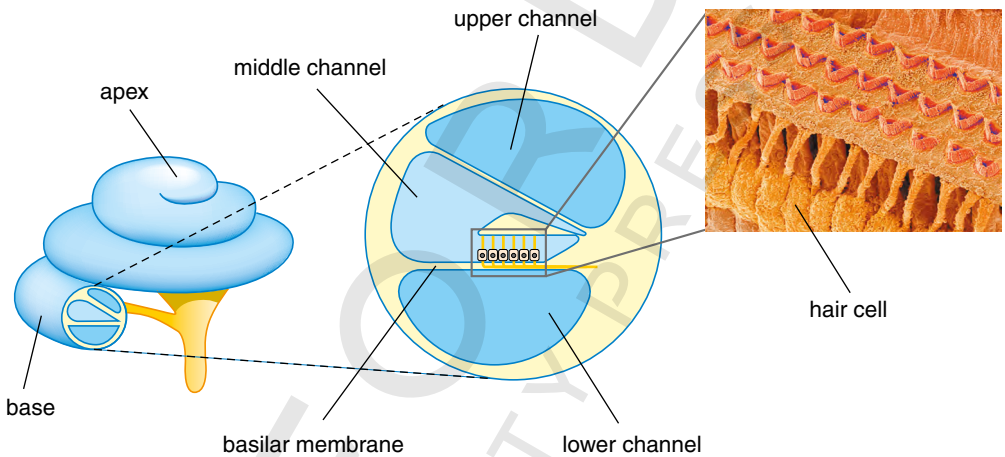


Fig 2.1e *Three channels in cochlea.*

✎ Damage to hair cells is a common cause of hearing loss. A common example is noise-induced hearing loss. Hair cells of high frequency (centred at 4000 Hz) sounds are typically damaged by noises. Hearing loss is mentioned on p.72.

When a sound wave enters the cochlea from the oval window, it propagates through the upper channel to the apex, and then through the lower channel to the round window. During this process, the **basilar membrane** in the middle channel is set into vibration (Fig 2.1f). When the basilar membrane vibrates, the hair cells on it are stimulated and send nerve pulses to the brain via the **auditory nerve**.

Note that the stiffness changes gradually along the basilar membrane. This makes different regions of the basilar membrane vibrate with peak amplitudes at different sound frequencies. From the position of the peak of stimulation, the brain can recognize the frequency of the incoming sound.

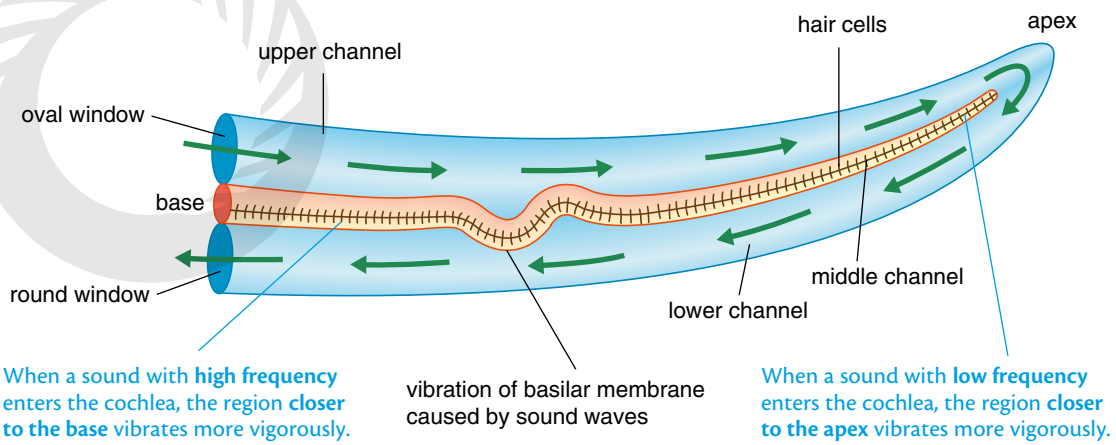


Fig 2.1f *How a sound wave propagates in the cochlea. The cochlea is imagined to be straightened.*

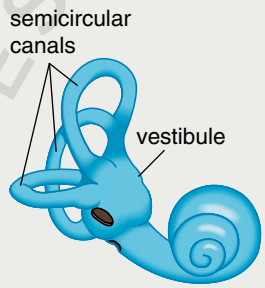
hair cell 毛細胞　basilar membrane 基底膜　auditory nerve 聽神經

The cochlea acts as a frequency analyser. Regions closer to its base vibrate with peak amplitudes for high frequency incoming sounds while regions closer to its apex vibrate with peak amplitudes for low frequency incoming sounds.

Know more

Inner ear for balance

Besides detecting sound, the ear also helps maintain body balance. This is achieved by the three semicircular canals and the vestibule in the inner ear. The canals are loops lying perpendicularly to each other and are connected to the vestibule. They can detect rotational and linear movements of our head and body. The sensory cells in them then generate signals to the brain. The brain integrates signals from the ears, eyes and muscles, then bring about responses to maintain body balance.



Google Forms Microsoft Forms

Quick check 2

→ E-version in OUP Exercise Platform is also available.

1 Which of the following is/are correct when sound of a certain frequency enters the cochlea? Put a tick in the corresponding box(es).

- ☐ The basilar membrane vibrates.
- ☐ Only the hair cells in a certain region vibrate.
- ☐ The hair cells in a certain region vibrate more vigorously than the others.

2 Which of the following statements about the cochlea is **incorrect**?

A It is fluid-filled.

B The basilar membrane has uniform stiffness.

C It can help in identifying sounds of different frequencies.

D The hair cells at different regions react differently to sounds of different frequencies.

Learn by practice 1

1 If the area of the ear drum is 18 times that of the oval window, and the force is increased by 1.2 times by the lever action of the ear bones, how many times is the increase in pressure?

2 If the force is increased by 1.42 times by the lever action of the ear bones, and the middle ear amplifies the pressure by 20 times, what is the ratio of the area of the oval window to the eardrum?

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DSE goal

Realize hearing perception of relative sound intensity levels.

Realize the need for logarithmic scale to represent the levels.

Apply $L = 10 \log \left(\frac{I}{I_0}\right)$ to solve problems.

DSE exam

13(2)Q4.5, 15(2)Q.4(c), 16(2)Q4.5, 17(2)Q4.3, 18(2)Q4.4, 19(2)Q4.3, 22(2)Q4.4

2 Loudness of sound

Loudness is the biological response to sound intensity. Let us study the idea of sound intensity first.

a Sound intensity

Sound waves carry energy. One way to describe the amount of energy carried by a sound wave is to use sound intensity I, which is defined as the amount of energy transmitted through each unit of area per unit time.

$$\text{Intensity} = \frac{\text{energy}}{\text{area} \times \text{time}} = \frac{\text{power}}{\text{area}}$$

(W m^{-2} for intensity; W for power; m^2 for area)

Consider a point source S which emits sound evenly in all directions. The sound spreads across a spherical surface centred at S (Fig 2.1g). As the sound from S travels further, it spreads across a larger spherical surface and the intensity decreases.

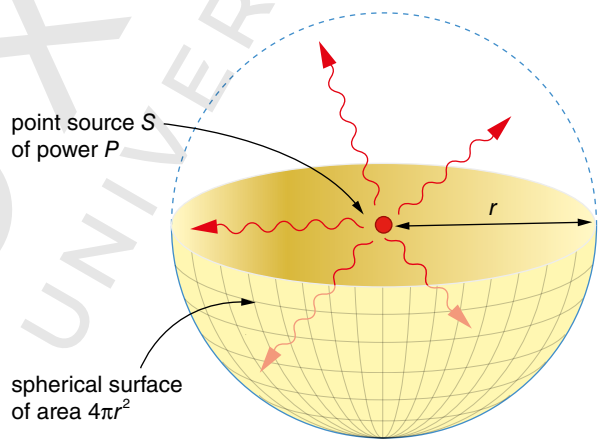


Fig 2.1g *Sound from a point source spreads evenly on a spherical surface.*

A spherical surface of radius r has an area of $4\pi r^2$. If the power of the source is P, the intensity at a distance r from the source will be

$$I = \frac{P}{4\pi r^2} \quad \text{(for point source)}$$

(I: **intensity** at a position (W m^{-2}); P: **power** of the source (W); r: **distance** between that position and the source (m))

Example 1 Sound intensity

S is a point source of sound. Position X is at 5 m from S. The sound intensity measured at X is 4.5×10^{-6} W m^{-2}. Find the power of the source.

Solution

By $I = \dfrac{P}{4\pi r^2}$,

$P = 4\pi r^2 I = 4\pi \times 5^2 \times (4.5 \times 10^{-6}) = 1.41 \times 10^{-3}$ W

Think about...

How will the intensity at X change if the power of the source is doubled?

Example 2 Sound intensity (using proportion)

S is a point source of sound. Positions X and Y are at d and $3d$ from S respectively. The sound intensity measured at X is 4.5×10^{-6} W m^{-2}. Find the sound intensity measured at Y.

Solution

By $I \propto \dfrac{1}{r^2}$,

$$\frac{I_Y}{I_X} = \frac{r_X^2}{r_Y^2} = \frac{d^2}{(3d)^2} = \frac{1}{9}$$

$$\therefore \quad I_Y = \frac{1}{9} \times I_X = \frac{1}{9} \times (4.5 \times 10^{-6}) = 5 \times 10^{-7} \text{ W m}^{-2}$$

Quick check 3

→ E-version in OUP Exercise Platform is also available.

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1 Complete the table below by applying $I = \dfrac{P}{4\pi r^2}$.

	I / W m^{-2}	P / W	r / m
(a)		5.6×10^{-3}	12
(b)	2.5×10^{-6}		20
(c)	6×10^{-6}	1.2×10^{-3}	

2 Given that $I = \dfrac{P}{4\pi r^2}$, which of the following is/are correct? Put a tick in the corresponding box(es).

☐ $I \propto P$ ☐ $I \propto r$ ☐ $I \propto r^2$

☐ $I \propto \dfrac{1}{P}$ ☐ $I \propto \dfrac{1}{r}$ ☐ $I \propto \dfrac{1}{r^2}$

Learn by practice 2

1 S is a point source of sound. The sound intensity measured at 3 m from S is 1.2×10^{-5} W m^{-2}. What will be the sound intensity at 10 m from S?

2 S is a point source of sound with power 2×10^{-3} W. The sound intensity measured at position X nearby is 8.4×10^{-6} W m^{-2}. If the power of the sound source is increased to 5×10^{-3} W, what will be the sound intensity measured at X?

3 S is a point source of sound. The sound intensity at a distance d from S is 4.0×10^{-5} W m^{-2}. What will be the sound intensity at a distance of $\frac{1}{2}d$ from S?

4 S is a point source of sound. The sound intensity measured at position Y nearby is 7.6×10^{-6} W m^{-2}. If the power of the sound source is halved, what will be the sound intensity measured at Y?

b Sound intensity level

The decibel scale was designed by taking advantage of logarithms and ratios. It is a logarithmic scaling system used to describe ratios with a very large range of values. It has a benefit of closely matching our perception of sound levels which is generally logarithmic.

For sound at 1000 Hz, the lowest intensity that a normal person can hear is about 10^{-12} W m^{-2} (represented by I_0) and the highest intensity that we can tolerate is about 100 W m^{-2}. Since this range of intensities is so large, it is more convenient to represent it in a **logarithmic scale**. We hence define a new quantity called **sound intensity level**. It is represented by L and the unit is **decibel** (**dB**). It is expressed as

$$L = 10 \log \left(\frac{I}{I_0} \right)$$

L: sound intensity level (dB); I: intensity of the sound (W m^{-2}); I_0: lowest intensity a normal person can hear at 1000 Hz = 10^{-12} W m^{-2}

Note that sound intensity level L compares a sound intensity I with a reference value I_0. It may have a negative value when I is smaller than I_0.

At 1000 Hz, the softest detectable sound corresponds to a sound intensity level of $10 \log \left(\frac{10^{-12}}{10^{-12}} \right) = 0$ dB, while the loudest tolerable sound corresponds to a sound intensity level of $10 \log \left(\frac{100}{10^{-12}} \right) = 140$ dB. We can see that this range of sound intensity level is much more manageable.

Cross-link
Refer to Book 3B Ch 7 for the sound intensity levels of different sounds in daily life.

A sound of 10 dB is just audible, e.g. whisper, a sound of 140 dB is at human ear pain threshold, e.g. a jet aircraft taking off. Normal conservation is at about 60 dB.

c Threshold of hearing

The **threshold of hearing** is the lowest level of sound that a person can hear. At 1000 Hz, it has an intensity of 10^{-12} W m^{-2} and a sound intensity level of 0 dB for a normal person.

sound intensity level 聲強級　decibel (dB) 分貝　threshold of hearing 聽覺閾

Example 3 Sound intensity and sound intensity level

Edward hears a sound from outside the window.
Given: threshold of hearing $I_0 = 10^{-12}$ W m^{-2}



(a) Suppose the sound reaching Edward has an intensity of 10^{-7} W m^{-2}. What is the sound intensity level as heard by Edward?

(b) Edward walks towards the window and stops next to it. If he hears a sound of 60 dB, what is the intensity of sound received by him?

Solution

(a) $L = 10 \log\left(\frac{I}{I_0}\right) = 10 \log\left(\frac{10^{-7}}{10^{-12}}\right) = 50$ dB

(b) $L = 10 \log\left(\frac{I}{I_0}\right)$

$60 = 10 \log\left(\frac{I}{10^{-12}}\right)$

$6 = \log\left(\frac{I}{10^{-12}}\right)$

$10^6 = \frac{I}{10^{-12}}$

$I = 10^{-6}$ W m^{-2}

→ Teaching notes and exercises are available on OUP web.

Skill *Mathematics*

Logarithm

▸ p.179

Quick check 4

→ E-version in OUP Exercise Platform is also available.

Google Forms | Microsoft Forms

1 Complete the table below by applying $L = 10 \log\left(\frac{I}{I_0}\right)$.

	L / dB	I / W	I_0 / W m^{-2}
(a)		2×10^{-2}	10^{-12}
(b)	46		10^{-12}
(c)		4.5×10^{-5}	10^{-12}
(d)	1		10^{-12}

2 A sound intensity level of 0 dB

A means no sound.

B is the lowest intensity level of sound that a person can hear at 1000 Hz.

C is the lowest intensity level of sound that a person can hear at any frequency.

☆ HKDSE 2013 Paper 2 Q4.5

→ This video demonstrates how this question is solved.

Video

DSE practice 1 | Power of source and sound intensity level

A speaker originally outputs sound with an average power of 10^{-4} W and Susan, staying in her seat, thinks that the sound is too soft. She increases the volume with a remote control so that the average power becomes 2×10^{-4} W. What is the increase in sound intensity level at Susan's position? (4 marks)

Solution

By $I \propto P$,

Before the change: P_1, I_1, L_1
After the change: P_2, I_2, L_2

$$\frac{I_2}{I_1} = \frac{P_2}{P_1} = \frac{2 \times 10^{-4}}{10^{-4}} = 2$$ 1M

$$L_1 = 10 \log \left(\frac{I_1}{I_0}\right) \text{ ..... (1)}$$ 1M

$$L_2 = 10 \log \left(\frac{I_2}{I_0}\right) \text{ ..... (2)}$$

Subtract (1) from (2):

$$L_2 - L_1 = 10 \left[\log \left(\frac{I_2}{I_0}\right) - \log \left(\frac{I_1}{I_0}\right)\right] = 10 \log \left(\frac{I_2}{I_0} \div \frac{I_1}{I_0}\right)$$

$$= 10 \log \left(\frac{I_2}{I_1}\right)$$ 1M

$$= 10 \log 2$$

$$= 3.01 \text{ dB}$$ 1A

As the exact values of sound intensities cannot be calculated, they have to be eliminated by using one of the laws of logarithms:

$\log a - \log b = \log (a \div b)$

∴ The sound intensity level at Susan's position increases by 3.01 dB.

▶ Revision 2 Q8 (p.107)

Learn by practice 3

1 A sound is produced by a drill at a construction site.

(a) If the intensity of sound detected from a certain distance is 2.8×10^{-3} W m^{-2}, what is the sound intensity level detected there?

(b) The sound intensity level heard by John outside the construction site is 90 dB. What is the intensity of sound reaching John?

2 Paul stays at a certain distance from a point source of sound. What is the change in the intensity level of sound reaching Paul if the power of the sound source is decreased from 5×10^{-3} W to 2×10^{-3} W?

3 A jet plane takes off and produces a sound. The intensity of the sound detected at a position X is 100 W m^{-2}. A staff working on the airport runway wears a pair of ear protectors. The ear protectors reduce the intensity of the sound reaching him to 3.6×10^{-5} W m^{-2} when he is at X. Find the reduction in the intensity level of the sound reaching the staff.

4 Janice hears a noise from road drilling outside the room. She then closes the window and the intensity level of the noise reaching her is reduced by 15 dB. Find the ratio of the intensity of the noise reaching Janice before and after closing the window.

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DSE goal

Interpret curves of equal loudness.

DSE exam

12(2)Q4.3, 14(2)Q4.2, 20(2)Q4.2, 23(2)Q4.2

Simulation

→ This simulation can check the relative sensitivity of your ears at different frequencies of sounds. It is available on OUP web in PC version (.exe) only.

✎ Sounds below and above the audible range are called infrasound and ultrasound respectively. Infrasound is used to monitor earthquakes. Animals like elephants and whales are known to use infrasound to communicate over distance. Ultrasound has vast variety of applications. The most common one is medical imaging. We shall study ultrasound in Chapters 2.2 and 2.3.

d Perceived loudness

The perceived loudness of sounds depends not only on sound intensity but also on frequency. A person may perceive sounds of the same intensity level but different frequencies as having different loudness. We can describe the relationship between the loudness and frequency by **curves of equal loudness** (Fig 2.1h).

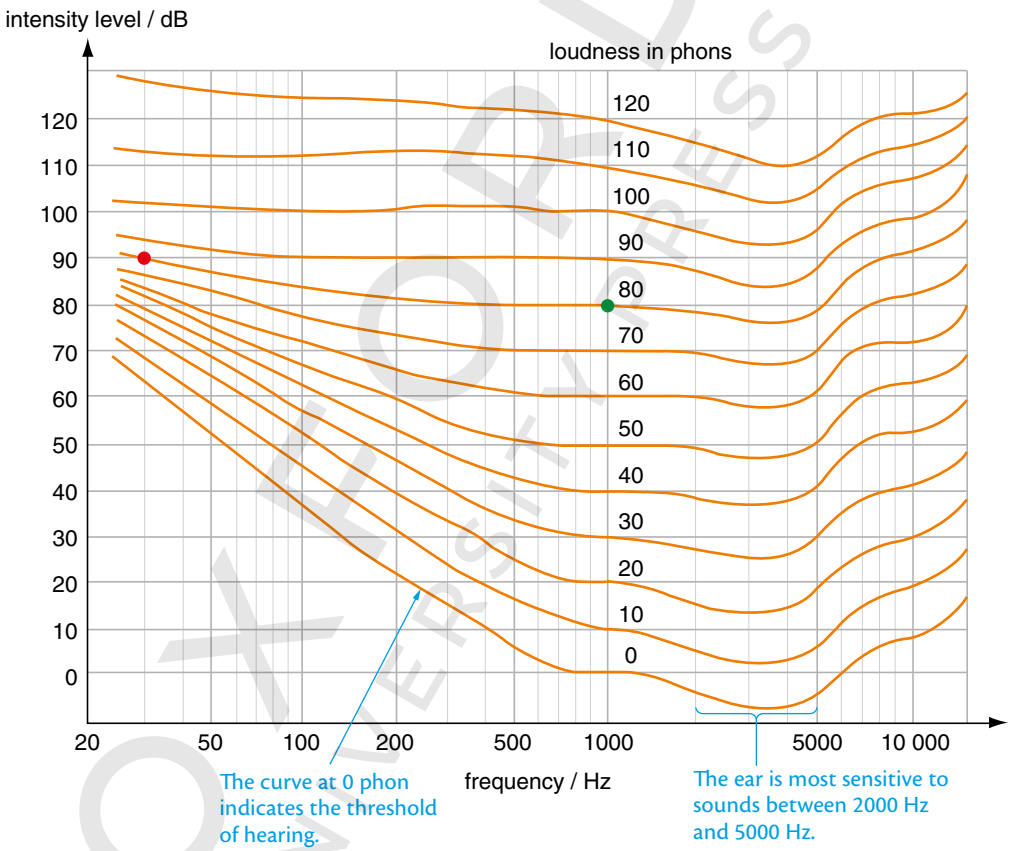


Fig 2.1h *Curves of equal loudness.*

Note the following:

1 Each curve represents the sound intensity levels that are perceived to have the same loudness at different frequencies.

For example, a sound of 90 dB and 30 Hz (red dot) has the same perceived loudness as a sound of 80 dB and 1000 Hz (green dot).

2 **Phon** is the unit of perceived loudness of sound. A sound of *n* phons has the same perceived loudness as a 1000-Hz pure note at *n* dB.

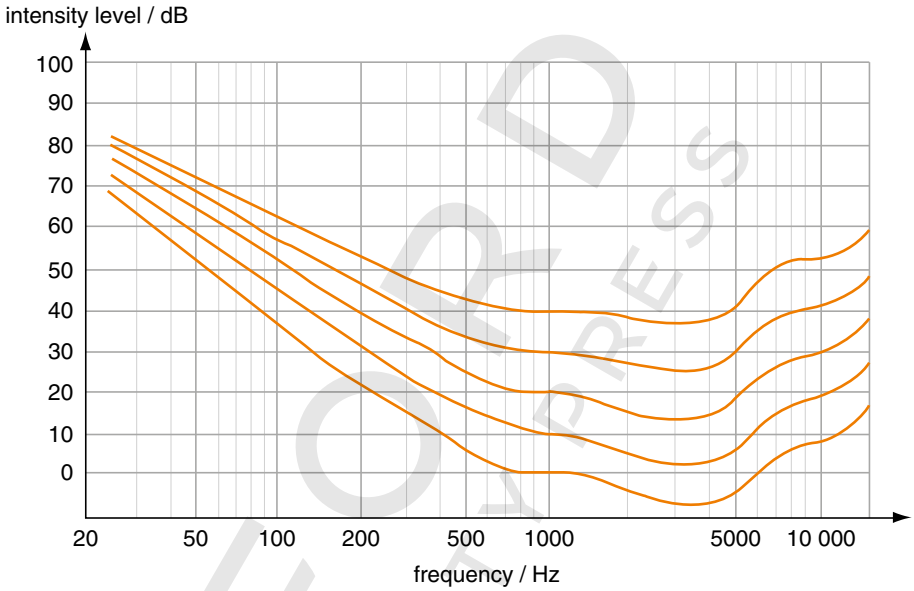
For example, a sound of 90 dB and 30 Hz (red dot) has a phon number of 80 because it has the same perceived loudness as a sound of 80 dB and 1000 Hz (green dot). The equal loudness curve which they lie on has a phon number of 80.

3 The threshold of hearing is represented by the equal loudness curve at 0 phon.

curve of equal loudness 等響曲線　　phon 方

Example 4 Curves of equal loudness

The following figure shows some curves of equal loudness of John.



(a) What is the phon number of a 200-Hz note at 40 dB?

(b) Compare the loudness of a 100-Hz note at 50 dB and a 10 000-Hz note at 40 dB as perceived by John.

(c) Which range of frequencies of sound is John's ear more sensitive to, 100–200 Hz or 1000–2000 Hz?

Solution

(a) 20

(b) The 10 000-Hz note at 40 dB is louder.

(c) 1000–2000 Hz

Tip
This is because the sound of 200 Hz and 40 dB lies on the curve which passes through the sound of 1000 Hz and 20 dB.

Tip
This is because the sound of 100 Hz and 50 dB is between 10 phons and 20 phons, while the sound of 10 000 Hz and 40 dB is between 20 phons and 30 phons.

Tip
This is because the part of an equal loudness curve within this frequency range is lower, meaning that sound within this frequency range needs a lower sound intensity level to provide the same perceived loudness.

Quick check 5

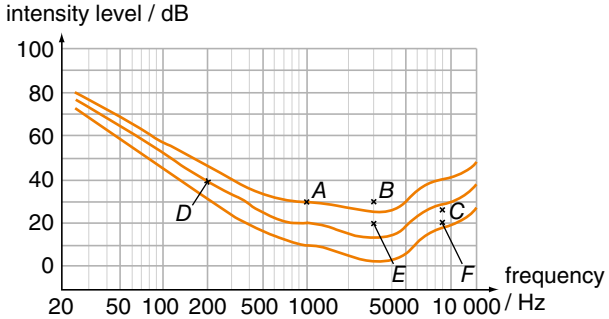
→ E-version in OUP Exercise Platform is also available.

Google Forms Microsoft Forms

1 The figure on the right shows some equal loudness curves of a person. *A*, *B*, *C*, *D*, *E*, *F* represent six different notes heard by the person.

(a) What are the phon numbers of notes *A* and *D*?

(b) Arrange the six notes in ascending order of loudness perceived by the person.



DSE goal

Discuss effects of noise on health of hearing.

DSE exam

15(2)Q.4(b), 18(2)Q4.3, 22(2)Q4.3

3 Effects of noise on health of hearing

Loud sound can damage hair cells in the cochlea. Damaged hair cells may fail to transmit electrical signals to the brain and result in hearing loss. Hearing loss may also be caused by aging. In either case, the person can only hear sound of higher intensity level. This is represented by an upward shift in the threshold of hearing. Figure 2.1i shows an example of how the threshold of hearing would be affected for a person with a hearing loss.

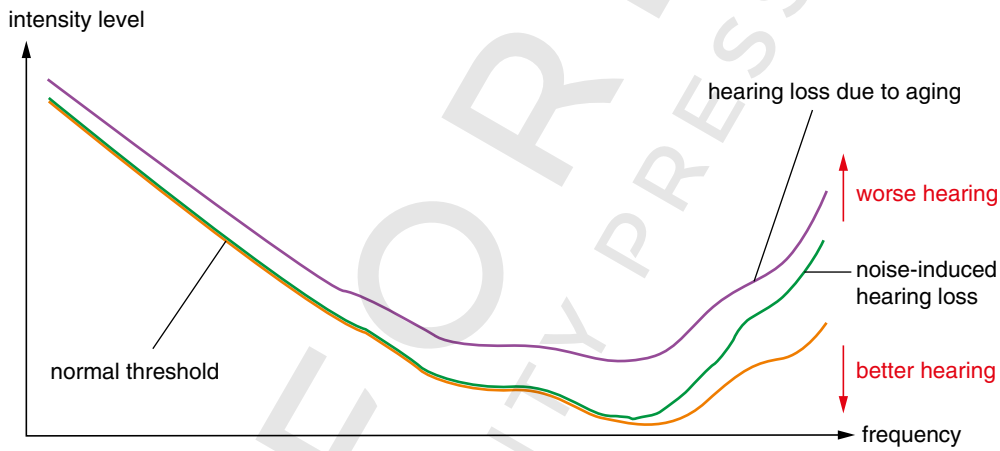


Fig 2.1i *Shift in threshold of hearing due to noise or aging.*

A hearing loss is represented by an upward shift in the threshold of hearing.

✎ The louder the noise is, the shorter the safe amount of exposure is, e.g. safe exposure at 85 dB is 8 hours but that of 91 dB is 2 hours.

Hearing loss can be temporary or permanent. For temporary hearing loss, the hearing sensitivity can return to normal after a short period of time. Permanent hearing loss may be caused by long duration or repeated exposure to loud noise, or a single loud noise above 140 dB. In this case, there is a permanent damage to hair cells and the hearing sensitivity will not return to normal.

Know more

Keep away from loud noise!

Listening to music with earbuds may lead to exposure of sounds reaching 115 dB or higher. It is generally advised to keep the volume below 60% the maximum volume and avoid listening to loud music for very long durations. Always wear ear protectors when working under loud noise condition. You may find more information in the World Health Organization website.

Useful link

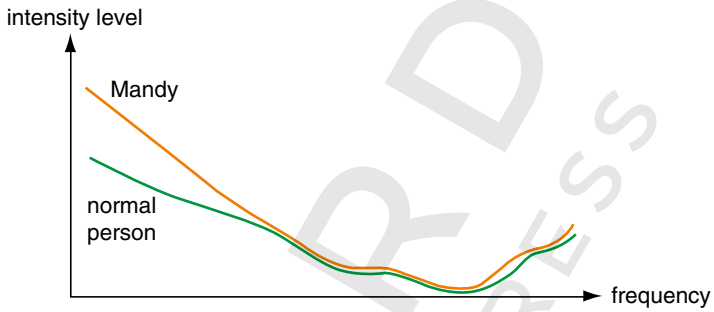
Hearing aids and electronic cochlear implants

People with mild to moderate hearing loss can hear sounds better by wearing hearing aids. If a person is deaf or has severe hearing loss, he/she may regain a sensation of hearing by having an electronic cochlear implant. Do you know how they work?

▸ p.114

Example 5 Hearing loss

The following figure shows the threshold of hearing of Mandy and a normal person.



(a) Is Mandy's hearing loss more severe at low pitch sound or high pitch sound?

(b) Explain your answer in (a).

Solution

(a) Low pitch sound

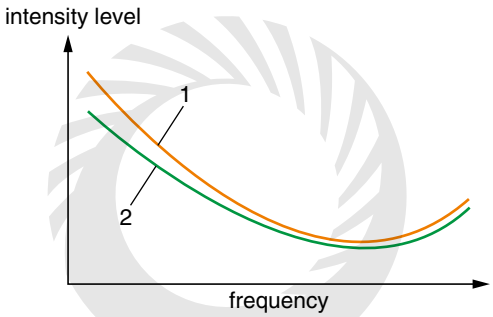
(b) The curve of Mandy is shifted upwards by a large amount from the curve of a normal person in the low frequency part, which means that the sound intensity level she needs to hear at low frequencies is much higher than a normal person.

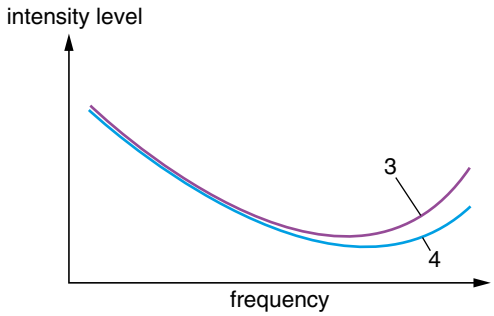
Quick check 6

→ E-version in OUP Exercise Platform is also available.

Google Forms Microsoft Forms

1 The figures below show the threshold of hearing of four different people *A*, *B*, *C* and *D*.





It is known that

i. one of the figures shows that *A* has a more severe hearing loss than *B* at low frequencies

ii. the other figure shows that *C* has a more severe hearing loss than *D* at high frequencies

Match *A*, *B*, *C* and *D* with their threshold of hearing.

A: _____; *B*: _____; *C*: _____; *D*: _____

Exercise 2.1

Unless otherwise specified, take $I_0 = 10^{-12}$ W m^{-2}.

1 Which of the following statements about hearing loss are correct?

(1) It can be caused by repeated exposure to loud noise.

(2) It can be caused by a single loud noise.

(3) The equal loudness curve for the threshold of hearing shifts downwards if there is a hearing loss.

A (1) and (2) only B (1) and (3) only

C (2) and (3) only D (1), (2) and (3)

***2** Patrick is listening to music emitted from a loudspeaker. The loudspeaker has a power of 5×10^{-5} W, and the sound intensity level received by Patrick is 45 dB. If the power of the loudspeaker is increased to 15×10^{-5} W, what is the sound intensity level received by Patrick?

A 48 dB B 50 dB

C 59 dB D 135 dB

***3** Which of the following statements about the threshold of hearing is/are correct?

(1) The intensity of sound at the threshold of hearing for a normal person is 0 W m^{-2} for a sound at 1000 Hz.

(2) It means that there is no sound.

(3) It refers to the sound at 0 phon.

A (3) only B (1) and (2) only

C (1) and (3) only D (1), (2) and (3)

***4** John hears a note of 1000 Hz and then a note of 50 Hz. He finds that the two notes are of equal loudness. It is known that John is more sensitive to sound at 1000 Hz than at 50 Hz. How are the phon number and the sound intensity level of the note of 1000 Hz compared to the note of 50 Hz?

	Phon number	Sound intensity level
A	same	lower
B	same	higher
C	lower	same
D	higher	same

***5** Describe the two pressure amplification mechanisms in the middle ear.

***6** A sound of a certain frequency enters the cochlea. How does the brain recognize its frequency?

***7** Describe how the hair cells closer to the base and the hair cells closer to the apex react differently when sounds of different frequencies enter the cochlea.

***8** S is a point source of sound. If the distance of Mary from S increases from 12 m to 20 m, what is the change in the intensity level of sound reaching Mary?

***9** A person stays at a certain distance from a point source of sound with power P. The power of the sound source is increased so that the intensity level of sound reaching the person increases by 5 dB. Find the new power of the sound source.

***10** Jessica is listening to music emitted from a loudspeaker. Figure a shows some curves of equal loudness of Jessica.

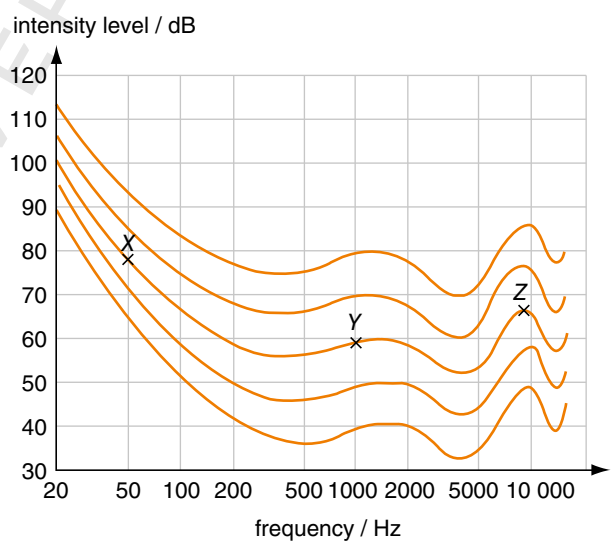


Fig a

(a) Jessica hears a sound of 70 phons at 50 Hz. Estimate its sound intensity level.

(b) Arrange the sounds X, Y and Z in ascending order of loudness perceived by Jessica.

(c) Jessica hears two sounds of 60 dB at 100 Hz and 200 Hz in turn. Which sound is louder as perceived by Jessica?

★**11** Susie is listening to music at home. Susie's father cannot hear the music clearly because he suffers from hearing loss due to aging. Figure b shows the threshold of hearing.

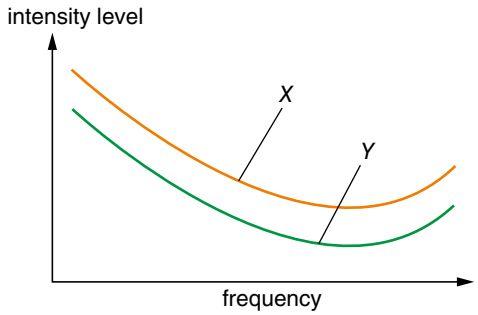


Fig b

Which curve represents the threshold of hearing of Susie's father? Explain briefly.

★**12** Figure c shows the structure of a human ear.

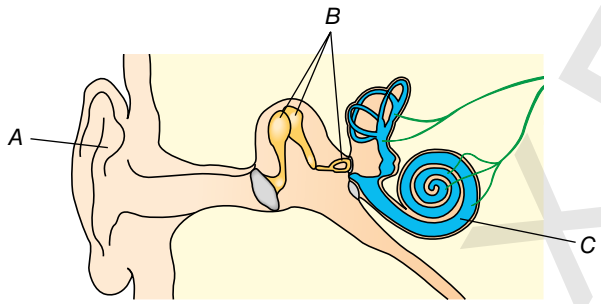


Fig c

(a) (i) Which structure, *A*, *B* or *C* amplifies the vibrations transmitted through it? Name that structure.

(ii) How does the structure in (a)(i) amplify the vibrations?

(b) (i) Which structure, *A*, *B* or *C* is responsible for frequency discrimination? Name that structure.

(ii) How does the structure in (b)(i) help in identifying sounds of different frequencies?

★**13** Figure d shows the threshold of hearing of a person suffering from a temporary hearing loss at high pitch sound.

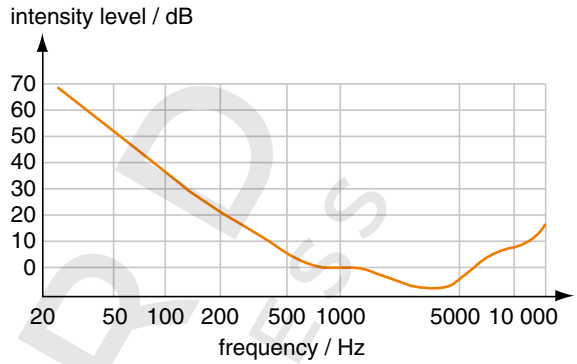


Fig d

(a) In the figure, shade the area which represents the sound that cannot be detected by the person.

(b) Sketch a possible curve for the threshold of hearing of this person when his hearing sensitivity returns to normal.

(c) Explain your answer in (b).

★**14** Cherry is watching television and the average intensity of sound at her position is 5×10^{-4} W m^{-2}.

(a) Find the sound intensity level received by Cherry.

(b) Cherry accidentally changes the volume of the television temporarily. The largest intensity of sound at her position becomes 10 times larger than the previous average value. State one possible effect on the hearing of Cherry.

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2.2 Ultrasound for medical imaging

Ultrasound scanning makes use of the **reflection of ultrasound** to produce image. It allows us to look inside our body. Examples include examining internal organs such as the heart and the liver, as well as monitoring a foetus during pregnancy. Like fibre optic endoscopy, ultrasound scanning is an imaging method that uses non-ionizing radiation.

In Book 3B Chapter 7, we learned about the properties of ultrasound. Let us have a review.

Video

→ This video shows an ultrasound scanner at work.

1 A review of ultrasound

Our audible frequency range is 20 Hz to 20 kHz. Ultrasound is a sound wave with a frequency higher than the upper limit of this range (Fig 2.2a), i.e. $f > 20$ kHz.

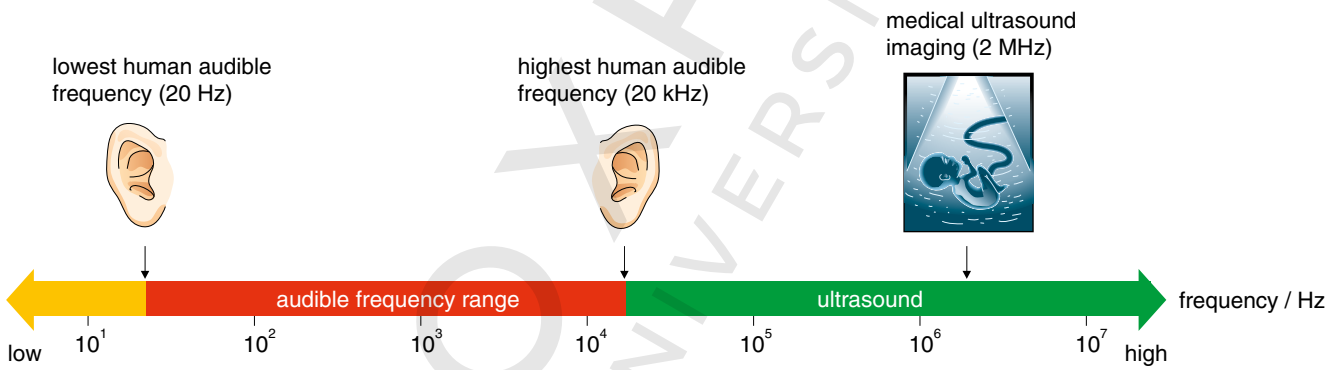


Fig 2.2a *Ultrasound in the sound frequency spectrum.*

Ultrasound has the same nature and properties as a sound wave:

1. It is produced by vibrations of particles.
2. It is a longitudinal wave: the particle vibrations are parallel to the direction the wave travels.
3. It requires a medium to travel.
4. It travels at different speeds in different media.
5. It obeys the wave equation $v = f\lambda$.
6. It shows wave properties of reflection, refraction, diffraction and interference.

In ultrasound scanning, we are mainly interested in the propagation and reflection of ultrasound. However, before going into the discussion on these properties, we will first look at how we can produce and detect ultrasound.

ultrasound scanning 超聲波掃描

2 Production and detection of ultrasound

Like audible sound, ultrasound is produced by mechanical vibrations. To produce vibrations at such high frequencies, we can make use of the oscillation of certain crystals.

DSE goal

Describe how piezoelectric transducer works.

DSE exam

14(2)Q.4(a)(i), 18(2)Q.4(a)

a Piezoelectric effect

Some crystals can generate a voltage when extended or compressed. This phenomenon is called the **piezoelectric effect**. The quartz shown in Figure 2.2b is a kind of piezoelectric crystal.

'Piezo-' is derived from the Greek word *piezein*, meaning 'to squeeze or press'.



Fig 2.2b *Quartz crystal, a kind of piezoelectric crystal.*

When a piezoelectric crystal is extended, a voltage is developed across the two sides. When the crystal is compressed, a voltage is developed in the opposite direction (Fig 2.2c).

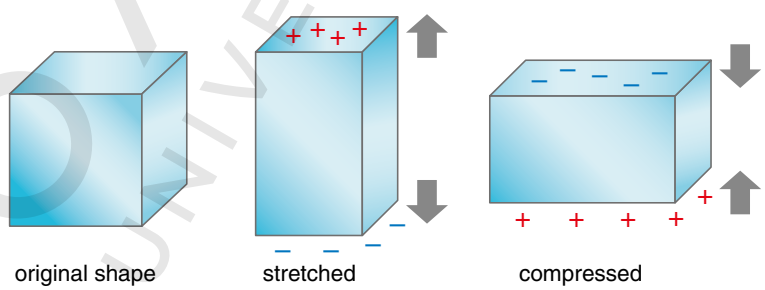


Fig 2.2c *Voltage developed due to the change of shape.*

This is called the reverse piezoelectric effect.

The piezoelectric effect is a reversible process. When a voltage is applied across a piezoelectric crystal, the crystal can become extended or compressed (Fig 2.2d).

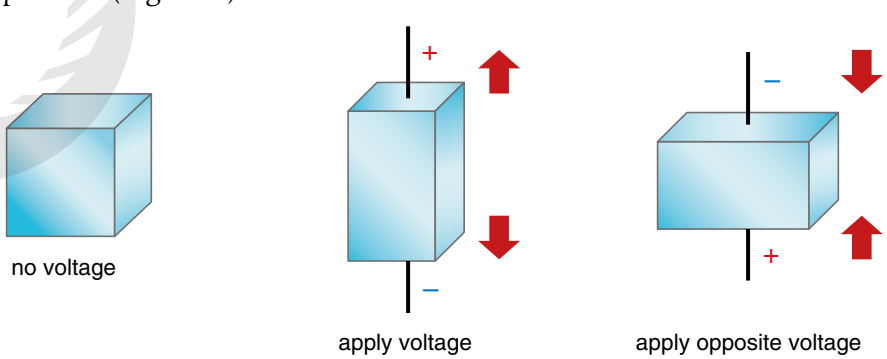


Fig 2.2d *Change in shape due to the voltage applied.*

piezoelectric effect 壓電效應

b Ultrasound transducer

In ultrasound scanning, ultrasound pulses are produced and detected by an ultrasound transducer (Fig 2.2e). An ultrasound transducer converts mechanical vibrations into electrical signals and vice versa.

An ultrasound transducer probe contains an array of piezoelectric crystals (Fig 2.2f) and makes use of the crystals to produce and detect ultrasound (Fig 2.2g). Note that the thickness of the piezoelectric crystal used in a transducer is not arbitrary.

This thickness depends on the desired frequency of the ultrasound.



Fig 2.2e *Ultrasound transducers of different sizes and shapes.*

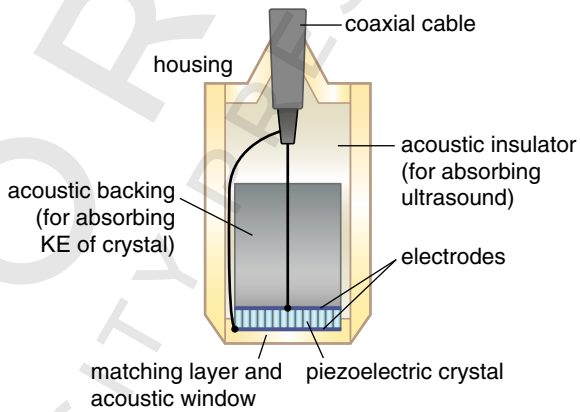


Fig 2.2f *Simplified internal structure of an ultrasound transducer.*

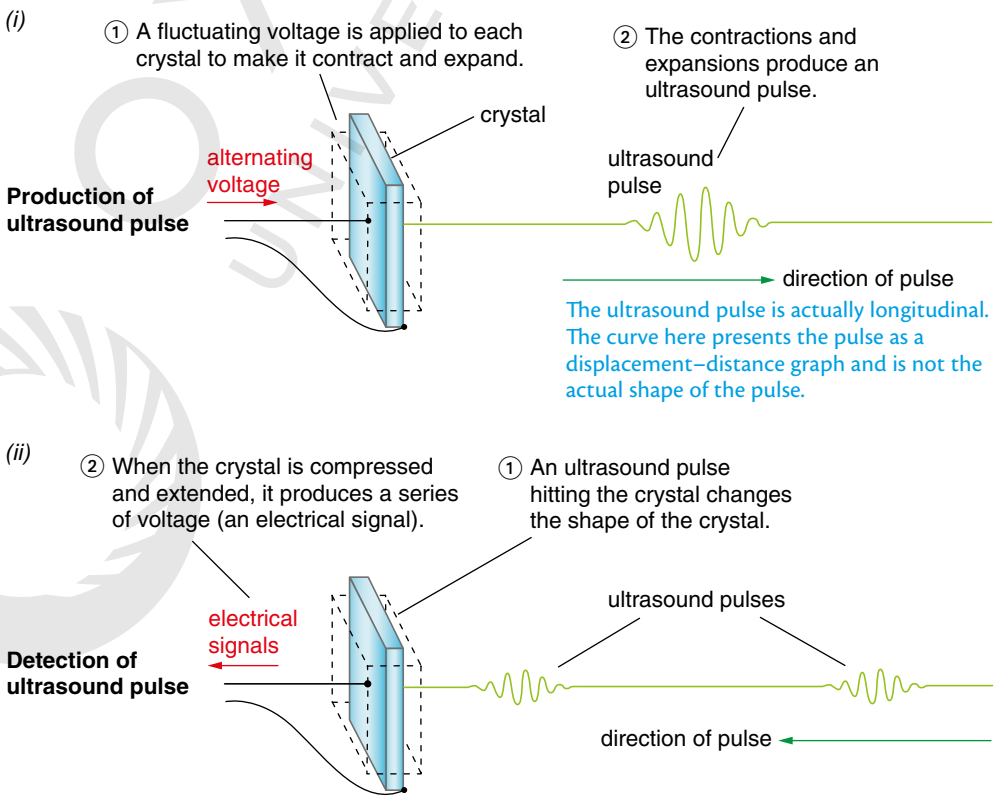


Fig 2.2g *(i) Production and (ii) detection of ultrasound pulse by a transducer.*

When a fluctuating voltage is applied across the piezoelectric crystal in an ultrasound transducer, the crystal contracts and expands to produce an ultrasound pulse.

When an ultrasound pulse hits the crystal, the crystal is compressed and extended to produce a series of voltage across it.

Google Forms Microsoft Forms

Quick check 7

→ E-version in OUP Exercise Platform is also available.

1 Which of the following frequencies is/are within the range of frequency of ultrasound? Put a tick in the corresponding box(es).

☐ 20 Hz ☐ 10 kHz ☐ 0.05 MHz

2 Arrange the following steps in a correct sequence to show how an ultrasound transducer detects an ultrasound pulse.

i The voltage is detected.

ii A series of voltage is developed in the crystal.

iii The ultrasound pulse reaches the piezoelectric crystal.

iv The crystal is stretched and compressed by the ultrasound pulse.

3 Arrange the following steps in a correct sequence to show how an ultrasound transducer produces an ultrasound pulse.

i Vibrations are produced in the medium as an ultrasound pulse.

ii A fluctuating voltage is applied to the crystal.

iii The piezoelectric crystal contracts and expands.

4 In which of the following can ultrasound propagate? Put a tick in the corresponding box(es).

☐ Bone ☐ Vacuum

☐ Water ☐ Air

Know more

Quartz clock

Quartz clocks are commonly found in our homes. Inside a quartz clock, a voltage is momentarily applied across a quartz crystal to make it expand and contract repeatedly at a known frequency (usually 32 768 Hz). The clock then can keep the time by a mechanism for counting the number of vibrations.



Natural oscillation of crystal

When you give a **single push** to a pendulum, it will keep on swinging back and forth for some time until it loses all its energy due to air resistance (Fig a). This is the natural oscillation of the pendulum. The frequency of this oscillation depends very much on the length of the string.

Similarly, when you **momentarily apply a voltage** to a piezoelectric crystal, it will contract and expand repeatedly for some time until it loses all its energy (Fig b). The frequency of this oscillation depends very much on the shape and size of the crystal.

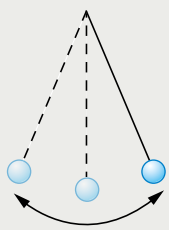


Fig a

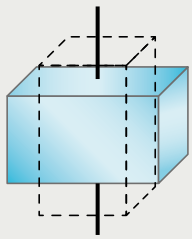


Fig b

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Know more

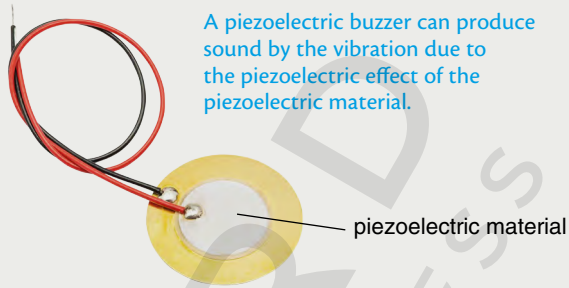
Other uses of piezoelectric effect

The piezoelectric effect is also applied in many different areas. The following are two examples.

A piezoelectric lighter can produce sparks by the high voltage due to the piezoelectric effect of the piezoelectric crystals inside.



A piezoelectric buzzer can produce sound by the vibration due to the piezoelectric effect of the piezoelectric material.



Can you find out more applications of the piezoelectric effect?

3 Reflection of ultrasound

Like other waves, ultrasound undergoes reflection when it reaches the boundary between two media. In an ultrasound scan, it is important to know how much energy is reflected and transmitted at the boundary. To find out these, we should first learn about a property of matter called **acoustic impedance**.

DSE goal

Define acoustic impedance.

Compare acoustic impedances of various body tissues.

a Acoustic impedance

For an ultrasound pulse travelling at a speed c in a medium with a density ρ, the acoustic impedance Z can be defined as

$$Z = \rho c$$

acoustic impedance of medium (kg m^{-2} s^{-1} or Rayl)

density of medium (kg m^{-3})

speed of sound in medium (m s^{-1})

The unit of acoustic impedance, Rayl, is named after *John William Strutt*, 3rd Baron Rayleigh.

Do not let the term 'impedance' confuse Ss. Used in the current context, it means ease of passage. 'Acoustic transparency' may have been a more intuitive term.

Table 2.2a lists the acoustic impedances of some body tissues and materials.

Medium	Z / 10^6 kg m^{-2} s^{-1}	Medium	Z / 10^6 kg m^{-2} s^{-1}
Air at 20 °C	0.000 441	Liver	1.66
Lung	0.26	Blood at 37 °C	1.66
Fat	1.38	Muscle	1.70
Water at 37 °C	1.52	Skin	1.99
Kidney	1.64	Bone (average)	6.50

Table 2.2a *Acoustic impedances Z of some body tissues and materials.*

acoustic impedance 聲阻抗　　John William Strutt 史特拉

Note that:

1 Most soft tissues have similar acoustic impedance values of about 1.6×10^6 kg m^{-2} s^{-1}.

This is because the lung contains a lot of air.

2 **Lung** is an exception. It has a **much lower** acoustic impedance value.

3 **Bone** has a **much higher** acoustic impedance value than soft tissues.

Google Forms Microsoft Forms

Quick check 8

→ E-version in OUP Exercise Platform is also available.

1 Complete the table below.

Medium	Acoustic impedance / kg m^{-2} s^{-1}	Density / kg m^{-3}	Speed of sound / m s^{-1}
A		1120	1530
B	1.66×10^6		1570
C	6.50×10^6	2030	

2 Which of the following body tissues has a much lower acoustic impedance than most of the other soft tissues?

A Bone
B Fat
C Lung
D Skin

3 Which of the following body tissues has a much higher acoustic impedance than soft tissues?

A Bone
B Fat
C Lung
D Skin

Learn by practice 4

1 Given that the density of the skin is 1150 kg m^{-3} and the speed of sound in the skin is 1730 m s^{-1}, find the acoustic impedance of the skin.

2 Given that the acoustic impedance of the heart is 1.65×10^6 kg m^{-2} s^{-1} and the density of the heart is 1060 kg m^{-3}, find the speed of sound in the heart.

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DSE goal

Apply intensity reflection coefficient to solve problems.

DSE exam

12(2)Q4.5, 17(2)Q4.4, 20(2)Q4.5, 22(2)Q4.5

b Intensity reflection coefficient

When an ultrasound pulse travels from one medium to another with a different acoustic impedance, part of the ultrasound is reflected and the remaining part is transmitted into the second medium (Fig 2.2h). Just like audible sound, the reflected ultrasound pulse is called an *echo*.

To describe the proportion of energy reflected, we can use the **intensity reflection coefficient** α, which is defined as

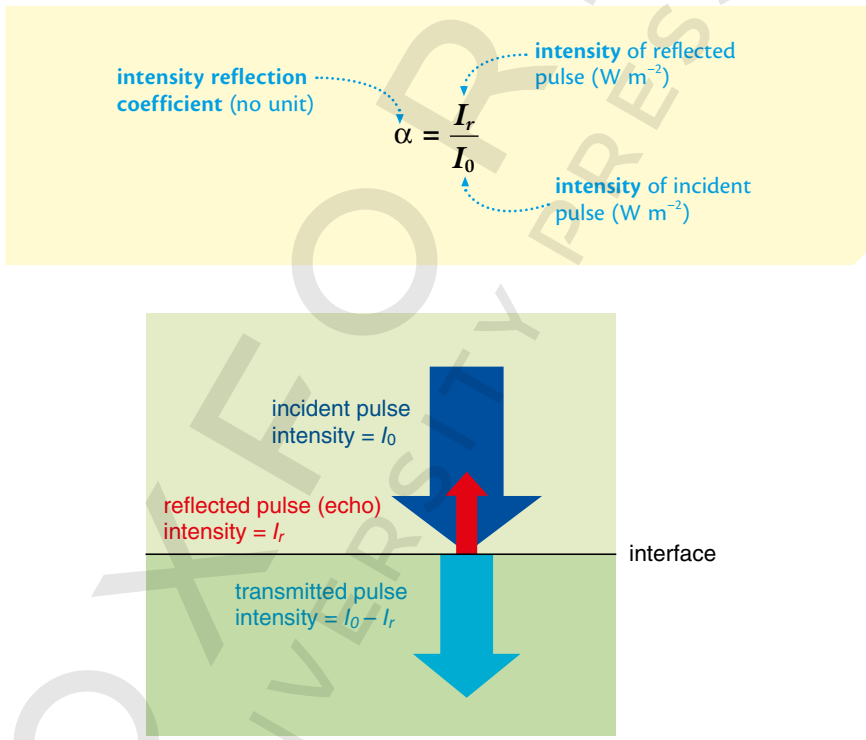
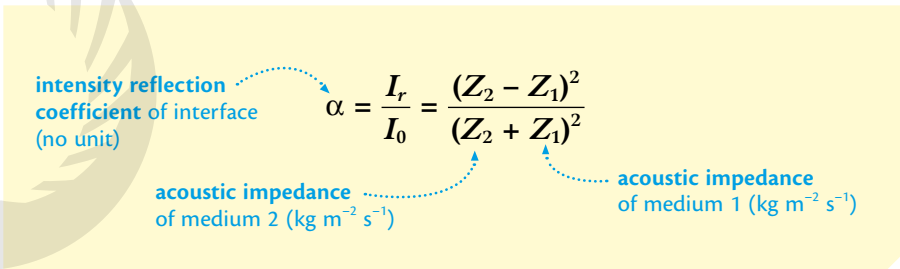


Fig 2.2h *Reflection and transmission of ultrasound pulse.*

Simulation

→ This simulation shows how the reflection and the transmission of ultrasound wave at an interface change with the acoustic impedances of the media. It is available on OUP web in PC version (.exe) only.

If an ultrasound pulse in a medium with acoustic impedance Z_1 enters normally into another medium with acoustic impedance Z_2, the intensity reflection coefficient α is given by:



At the extremes, if $Z_1 \approx Z_2$, then $\alpha \approx 0$, which means nearly no reflection occurs and all of the pulse is transmitted. If $Z_1 << Z_2$ or $Z_1 >> Z_2$, then $\alpha \approx 1$, which means most of the pulse is reflected.

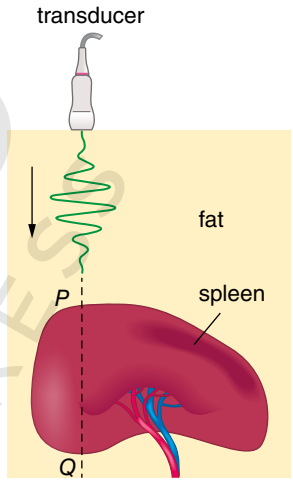
From this equation, we can see that if the difference in acoustic impedances is large, a **large proportion of the pulse will be reflected**. This happens when the pulse meets the interface between the **lung and other soft tissues**, or the interface between a **bone and soft tissues**.

Refer to points 2 and 3 on p.81. ▶

echo 回聲 / 回波　　intensity reflection coefficient 反射聲強係數

Example 7 Proportion of energy reflected and transmitted

An ultrasound pulse travels towards the spleen as shown. The acoustic impedances of fat and spleen are 1.38×10^6 kg m^{-2} s^{-1} and 1.65×10^6 kg m^{-2} s^{-1} respectively.



(a) Find the intensity reflection coefficient at interface *P*.

(b) Find the proportion of energy of the pulse (i) reflected, (ii) transmitted if the pulse is incident on interface *P* normally.

Solution

(a) $\alpha = \dfrac{(Z_2 - Z_1)^2}{(Z_2 + Z_1)^2} = \dfrac{(1.65 \times 10^6 - 1.38 \times 10^6)^2}{(1.65 \times 10^6 + 1.38 \times 10^6)^2} = 0.007\ 94$

(b) (i) Proportion of energy reflected = $\alpha = 0.007\ 94$

(ii) Proportion of energy transmitted = $1 - \alpha = 1 - 0.007\ 94 = 0.992\ 06$

Think about...

1 What is the intensity reflection coefficient at interface *Q*?

2 If the intensity of ultrasound pulse incident on interface *P* is 100 W m^{-2}, what are the intensities of the reflected pulse and the transmitted pulse?

By substituting
$Z_1 = 0.000\ 441 \times 10^6$ kg m^{-2} s^{-1}
$Z_2 = 1.99 \times 10^6$ kg m^{-2} s^{-1}
into $\alpha = \dfrac{(Z_2 - Z_1)^2}{(Z_2 + Z_1)^2}$

Cross-link

This explains why it is stated on p.61 that if sound waves directly transmit from the air-filled outer ear to the fluid-filled inner ear, only about 0.1% of the sound energy can be transferred through while most of the energy will be reflected back.

▶ From Table 2.2a on p.80 we can calculate that $\alpha = 0.999$ for an air-skin interface. Therefore, if there is any air between the ultrasound transducer and the skin, most of the ultrasound will be reflected and cannot enter the body (Fig 2.2i).

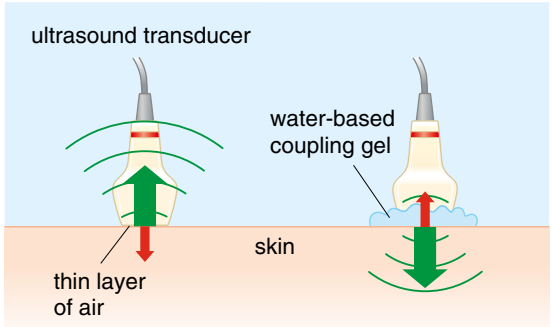


Fig 2.2i *Using ultrasound transducer without and with coupling gel.*

This is similar to the situation where wearing swimming goggles under water (see p.8). In both cases, air and gel are the coupling media. They change the relative amount of refracted and reflected waves so as to improve the wave transmission into another medium.



Fig 2.2j *Coupling gel is applied to the scan area.*

Google Forms Microsoft Forms

To solve this problem, a layer of water-based **coupling gel** is applied to eliminate any air between the skin and the transducer (Fig 2.2j). The gel has an acoustic impedance similar to that of the soft tissues and allows most of the ultrasound to pass into the body.

Quick check 9

→ E-version in OUP Exercise Platform is also available.

1 Complete the table below. Z_1 and Z_2 denote the acoustic impedances of media 1 and 2 respectively. Assume $Z_1 < Z_2$.

	Intensity reflection coefficient	Z_1 / kg m^{-2} s^{-1}	Z_2 / kg m^{-2} s^{-1}
(a)		1.38×10^6	1.64×10^6
(b)	0.724		1.66×10^6
(c)	0.958	1.99×10^6	

2 *True or false:* For the interface in Q1(a), if two ultrasound pulses of the same intensity travel from media 1 to 2 and from media 2 to 1 respectively, the intensity of the reflected pulses will be equal. (T/F)

3 Three identical ultrasound pulses hit normally on the three interfaces in Q1.

(a) In which case will the largest proportion of energy be reflected?

(b) In which case will the largest proportion of energy be transmitted?

4 In ultrasound scanning, water-based coupling gel is applied on the skin to

- ☐ eliminate the air between the skin and the transducer.
- ☐ provide a medium with an acoustic impedance very close to 0.
- ☐ provide a medium with an acoustic impedance close to that of the skin.
- ☐ provide a medium with a very large acoustic impedance.

Learn by practice 5

1 The acoustic impedances of air and skin are 411 kg m^{-2} s^{-1} and 1.99×10^6 kg m^{-2} s^{-1} respectively. Find the intensity reflection coefficient between air and skin.

2 The acoustic impedances of fat and muscle are 1.38×10^6 kg m^{-2} s^{-1} and 1.70×10^6 kg m^{-2} s^{-1} respectively. Find the intensity reflection coefficient between fat and muscle.

3 An ultrasound pulse travels from medium 1 to medium 2. The intensity reflection coefficient of the interface is 0.04.

(a) If the acoustic impedance of media 1 is 40 kg m^{-2} s^{-1}, find the possible acoustic impedance of media 2.

(b) The intensity of the pulse incident on the interface is 20 W m^{-2}.

(i) Find the reflected intensity.

(ii) Find the transmitted intensity.

coupling gel 耦合凝膠

Exercise 2.2

(For Q1–2.) The table below shows the density ρ of media P, Q, R and S and the speed c of ultrasound in them (Table a).

Medium	ρ / kg m^{-3}	c / m s^{-1}
P	1.3	330
Q	925	1590
R	1080	1450
S	1750	4080

Table a

★**1** Arrange the four media in ascending order of their acoustic impedances.

A $P < Q < R < S$

B $P < R < Q < S$

C $P < S < Q < R$

D $R < Q < S < P$

★**2** At which of the following interfaces is ultrasound reflected the least?

A P-Q interface B P-S interface

C Q-R interface D R-S interface

★**3** Which of the following statements about ultrasound are correct?

(1) Ultrasound has a frequency below 20 Hz.

(2) We cannot hear ultrasound.

(3) Ultrasound is a longitudinal wave.

A (1) and (2) only B (1) and (3) only

C (2) and (3) only D (1), (2) and (3)

★**4** An ultrasound pulse travels from one medium with acoustic impedance Z_1 to another medium with acoustic impedance Z_2. In which of the following cases will most of the ultrasound be reflected?

(1) Z_1 is much smaller than Z_2.

(2) Z_1 is about the same as Z_2.

(3) Z_1 is much larger than Z_2.

A (1) only B (2) only

C (3) only D (1) and (3) only

5 (a) State the range of frequency of sound that is detectable by a normal person.

(b) State the definition of ultrasound.

6 Briefly describe what happens when

(a) a constant voltage is applied across a piezoelectric crystal.

(b) a piezoelectric crystal is extended or compressed.

7 Explain how an ultrasound transducer produces an ultrasound pulse.

8 Explain how an ultrasound transducer detects an ultrasound pulse reaching it.

★**9** A beam of ultrasound of frequency 2 MHz travels through a liquid. The wavelength of ultrasound in the liquid is 7.2×10^{-4} m and the acoustic impedance of the liquid is 1.29×10^{6} kg m^{-2} s^{-1}.

(a) What is the speed of ultrasound in the liquid?

(b) What is the density of the liquid?

★**10** The acoustic impedances of air, water and soft tissue are 400 kg m^{-2} s^{-1}, 1.48×10^{6} kg m^{-2} s^{-1} and 1.63×10^{6} kg m^{-2} s^{-1} respectively.

(a) What is the intensity reflection coefficient

(i) between water and soft tissue?

(ii) between air and soft tissue?

(b) Explain why a layer of water-based coupling gel is applied between the transducer and the skin in ultrasound imaging.

★**11** An ultrasound pulse travels from coupling gel to skin. About 96.8% of energy of the incident pulse is transmitted through the interface between the coupling gel and the skin.

(a) What is the frequency range of ultrasound?

(b) Find the percentage of energy reflected by the interface.

(c) The acoustic impedance of skin is 1.99×10^{6} kg m^{-2} s^{-1}. What is the acoustic impedance of the coupling gel if it is smaller than this value?

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2.3 Ultrasound scanning

Have you ever seen an ultrasound image of a foetus? Here is a snapshot of a 2-D ultrasound image of a foetus (Fig 2.3a). Do you know how is this image produced? Why does the spine appear bright in the image?

Refer to p.93.

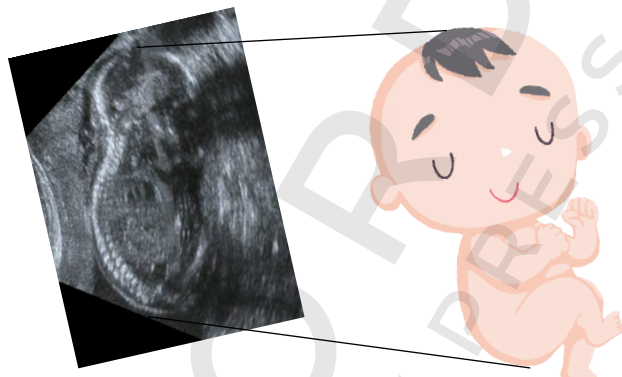
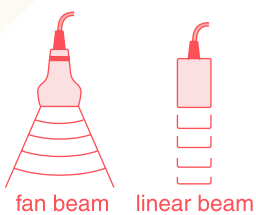


Fig 2.3a Ultrasound image of a foetus.

The left part of the image is the body surface. The image here is produced by a curvilinear transducer which sends out a fan beam and produces a sector type image, allowing a wider field of view.



DSE goal

Realize ultrasound scanning as range-measuring system.

DSE exam

18(2)Q.4(c)

1 Range measurement by ultrasound

In an ultrasound scan, a transducer produces an ultrasound pulse which is transmitted into the patient's body. As we learned in Chapter 2.2, when the pulse reaches the interface between two different body structures, part of it is reflected and the other part continues to transmit to the second structure. The transducer detects the reflected pulses (echoes) and converts them into electrical signals (Fig 2.3b).

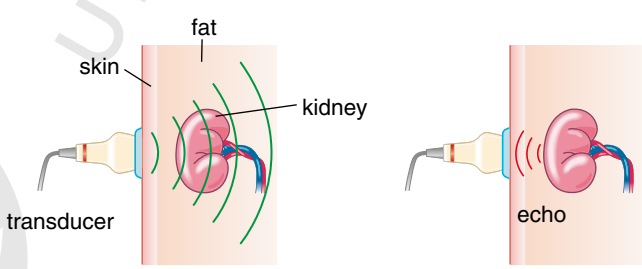


Fig 2.3b Ultrasound scanning obtains images by the reflection of ultrasound.

As the time taken by an ultrasound pulse to travel a round trip to an interface is measured (Fig 2.3c), the distance of the interface from the transducer can be calculated.

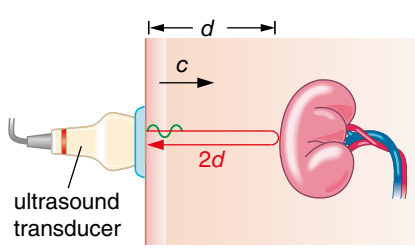


Fig 2.3c An ultrasound pulse travels a round trip before being detected.

In Figure 2.3c on p.86, the ultrasound pulse is sent into the body and travels at speed c. After some time t, the transducer detects a pulse reflected from the surface of the kidney.

If the distance between the transducer and the surface of the kidney is d, the detected pulse actually takes time t to travel a total distance of $2d$ in a round trip. Therefore, we have

$$2d = c \times t$$

$$d = \frac{1}{2}ct$$

Now consider a pulse being reflected from the front and the back surfaces of a structure (Fig 2.3d).

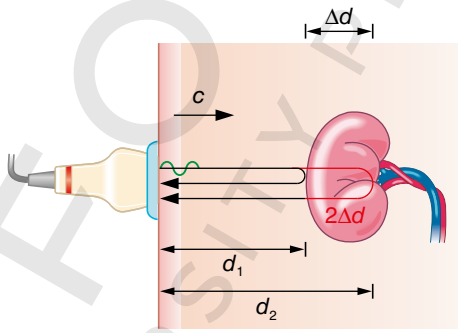


Fig 2.3d *An ultrasound pulse is reflected from the front and back surfaces of a kidney.*

If the front and back surfaces are at distances d_1 and d_2 from the transducer, and the reflected pulse 1 from the front surface and the reflected pulse 2 from the back surface are detected at time t_1 and t_2 respectively, reflected pulse 2 actually travels an **extra distance** of $\mathbf{2\Delta}\boldsymbol{d}$ with an **extra time** $\boldsymbol{\Delta t}$ within the structure, where

$$\Delta d = d_2 - d_1 \text{ and } \Delta t = t_2 - t_1$$

Note that Δd is the thickness of the structure along the travelling direction of the pulse. If the ultrasound pulse has a **speed c within the structure**, we have

$$2\Delta d = c \times \Delta t$$

This formula applies even the speeds of the pulse are different in other structures it has passed. ▶

$$\Delta \boldsymbol{d} = \frac{\mathbf{1}}{\mathbf{2}} \boldsymbol{c} \Delta \boldsymbol{t}$$

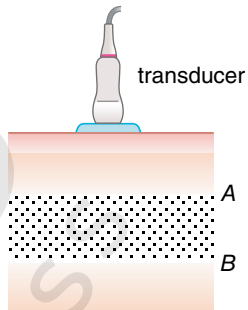
speed of pulse in the structure (m s^{-1})

thickness of structure (m)

time difference in receiving the two echoes (s)

Example 8 | Depth and thickness

An ultrasound pulse is emitted by the transducer at time $t = 0$ as shown. The echoes from interfaces A and B are detected by the transducer at $t = 15.4$ μs and 21.2 μs respectively. Assume that the speed of ultrasound along its path is 1550 m s^{-1}.



(a) Find the depth (in mm) of interface A from the transducer.

(b) Find the thickness (in mm) of the structure between interfaces A and B.

Solution

(a) $d = \frac{1}{2}ct = \frac{1}{2} \times 1550 \times (15.4 \times 10^{-6}) = 0.0119 \text{ m} = 11.9 \text{ mm}$

(b) $\Delta d = \frac{1}{2}c\Delta t = \frac{1}{2} \times 1550 \times (21.2 \times 10^{-6} - 15.4 \times 10^{-6}) = 0.004\,50 \text{ m} = 4.50 \text{ mm}$

Google Forms Microsoft Forms

Quick check 10

→ E-version in OUP Exercise Platform is also available.

1 In an ultrasound scan, the thickness of the body structure is Δd, the times for the transducer to receive the echo reflected from the front and the back surfaces of the structure are t_1 and t_2 respectively, the speed of the ultrasound in the structure is c. Complete the table below.

	Δd	c	t_1	t_2
(a)		1550 m s^{-1}	18.2 μs	20.5 μs
(b)		1450 m s^{-1}	5.2 μs	10.9 μs
(c)	2.3 mm		8.6 μs	11.6 μs

2 In computing the thickness of a structure using $\Delta d = \frac{1}{2}c\Delta t$, c should be equal to the

A average speed of the ultrasound pulse during its complete travel.

B speed of the ultrasound pulse before it enters the structure.

C speed of the ultrasound pulse within the structure.

Learn by practice 6

1 A transducer sends an ultrasound pulse towards an organ. Echoes from the front and the back surfaces of the organ are respectively received 25 μs and 47 μs after the emission of the pulse. Find

(a) the thickness of the organ, and

(b) the distance of the front surface of the organ from the transducer along the travelling direction of the pulse.

Take the speed of sound in the body be 1540 m s^{-1}.

4 A-scan and B-scan

There are different types of ultrasound scans. We will introduce two of them, **A-scan** and **B-scan**, each using a different display method.

DSE goal

Describe how A-scan works.

Interpret pulse display of A-scan.

DSE exam

12(2)Q.4(a), 14(2)Q4.3, 15(2)Q4.4, 17(2)Q4.5, 19(2)Q.4(b)

Simulation

→ This simulation shows the formation of an A-scan display. It is available on OUP web in PC version (.exe) only.

a A-scan

i Production of an A-scan image

In A-scan, a narrow beam of ultrasound pulse is used. The transducer output is connected to an oscilloscope (Fig 2.3e) and a one-dimensional image is produced.

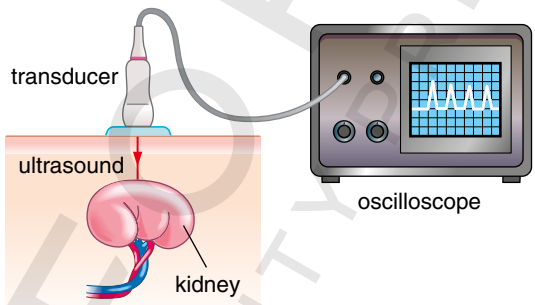


Fig 2.3e *Performing A-scan.*

The production of an A-scan image can be summarized as follows:

1. A transducer emits a narrow beam of ultrasound pulse into the body.
2. The transducer detects the echoes from different interfaces at different times.
3. Each echo is plotted as a spike on the A-scan image (Fig 2.3f). The **horizontal position** of a spike indicates the **time of detection** and the **amplitude** of the spike indicates the signal strength, which is proportional to the **intensity of the echo**.

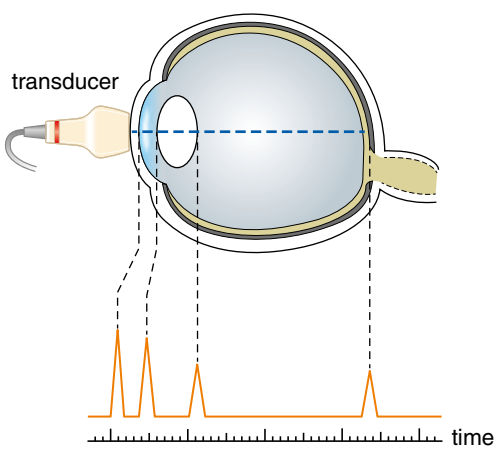


Fig 2.3f *The spikes on an A-scan image represent the echoes from different interfaces.*

The amplitude of a spike indicates the intensity of an echo.

▶ Note that the 'A' in **A**-scan stands for **amplitude**.

A-scan A-掃描　　B-scan B-掃描

ii Interpreting an A-scan image

The spikes in an A-scan image tell us about the time of detection of echoes and their intensities.

The **time of detection** of an echo depends on the speed of the ultrasound in the body and the depth (position) of the interface. If the speed is known, the **depth of the interface** can then be calculated.

Cross-link
See p.87 to recall how to calculate the depth of an interface by using the 'round trip time' of an ultrasound pulse.

The **intensity** of an echo depends mainly on the **intensity reflection coefficients of all the interfaces** the echo has reached. An echo with high intensity is usually caused by an interface with high intensity reflection coefficient. It could be the surface of lung or bone, or the surface of object caused by disease, e.g. gallstone or kidney stone.

Example 9 Finding thickness of a structure

A patient is having an ultrasound scan for his eye (Fig a). Figure b shows the A-scan image obtained.

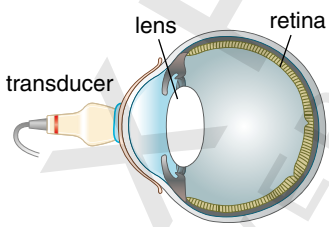


Fig a

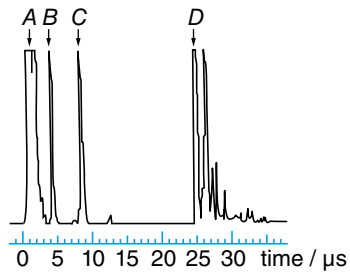


Fig b

Assume that the speed of ultrasound in the eye is 1550 m s^{-1}.

(a) Spike *D* corresponds to the retina. Find the distance (in mm) of the retina from the transducer.

(b) (i) Which spike corresponds to the front surface of the lens? Which spike corresponds to the back surface?

(ii) Find the thickness of the lens (in mm).

Solution

(a) $d = \frac{1}{2}ct = \frac{1}{2} \times 1550 \times (25 \times 10^{-6}) = 0.0194 \text{ m} = 19.4 \text{ mm}$

(b) (i) Spike *B* corresponds to the front surface.
Spike *C* corresponds to the back surface.

(ii) $\Delta d = \frac{1}{2}c\Delta t = \frac{1}{2} \times 1550 \times (8 \times 10^{-6} - 4 \times 10^{-6}) = 0.0031 \text{ m} = 3.1 \text{ mm}$

Think about...

How will the answer in (a) and (b)(ii) change if the speed of ultrasound in the lens is 1500 m s^{-1} (still 1550 m s^{-1} in other parts)?

Example 10 Intensity of echoes in A-scan image

A narrow beam of ultrasound pulse is emitted from the transducer to perform an A-scan (Fig a). Spikes *M* and *N* correspond to the detection of echoes from interfaces *P* and *Q* respectively.

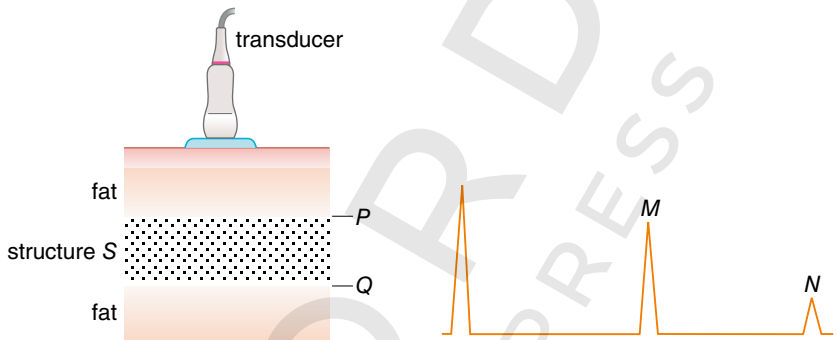


Fig a

(a) Spike *M* has a large amplitude. What can you tell about the intensity reflection coefficient of interface *P*? Explain.

(b) Spike *N* has a much smaller amplitude. Why?

Solution

(a) The large amplitude of spike *M* indicates a high intensity of the echo from interface *P*, and thus a large intensity reflection coefficient of this interface.

(b) Interface *P* reflects nearly all of the ultrasound.

Think about...

1 What can you tell about the acoustic impedance of structure *S*?

2 Does the small amplitude of spike *N* indicate a small intensity reflection coefficient of interface *Q*?

Google Forms Microsoft Forms

Quick check 11

→ E-version in OUP Exercise Platform is also available.

1 Arrange the following steps in producing a spike on the A-scan image in correct order.

i The transducer receives the echo.
ii A spike corresponding to the interface is displayed on the image.
iii The transducer emits an ultrasound pulse.
iv The transducer produces an electrical signal.
v The pulse is reflected by an interface.

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2 The horizontal position of a spike on an A-scan image indicates the (intensity / time of detection) of an echo from (a medium / an interface).

3 Which of the following graphs best represents the A-scan display of the body structure shown in Figure a?

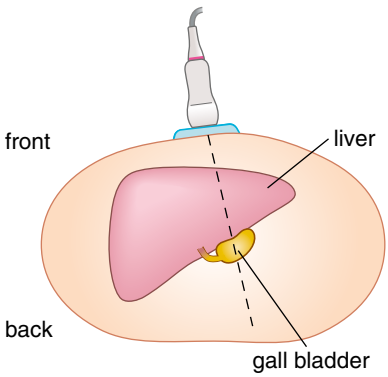
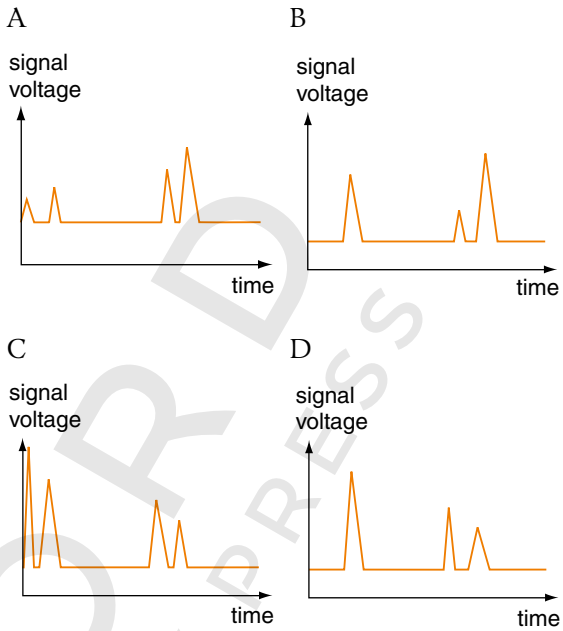


Fig a



Learn by practice 7

1 A patient's kidney is examined using ultrasound. Figure a shows the A-scan display. Spikes *X* and *Y* indicate the echoes from the skin surface and the kidney surface respectively.

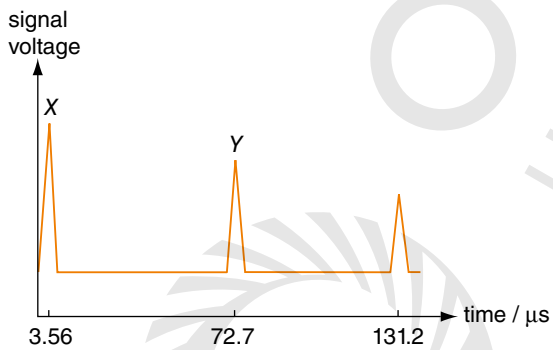


Fig a

Take the speed of ultrasound be 1540 m s^{-1}.

(a) Find the distance of the kidney surface from the transducer.

(b) Find the depth of the kidney under the skin.

2 A kidney is examined using ultrasound (Fig b). Figure c shows the A-scan display.

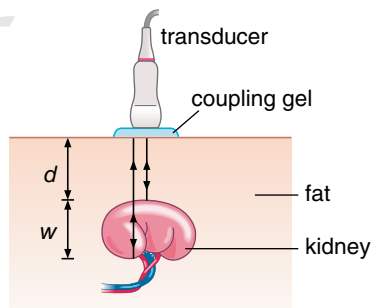


Fig b

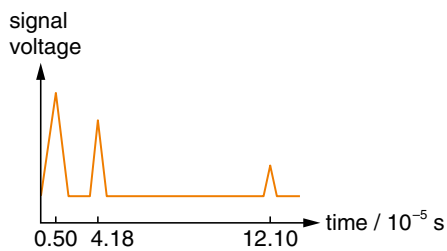


Fig c

Assume the speed of sound in skin and fat is 1460 m s^{-1} and that in kidney is 1540 m s^{-1}.

(a) Find the depth *d* of the kidney under the skin.

(b) Find the thickness *w* of the kidney.

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DSE goal

Describe how B-scan works.

Estimate size of body tissue in B-scan image.

DSE exam

12(2)Q4.4, 12(2)Q.4(b)(i), 15(2)Q4.3

Simulation

→ This simulation shows the formation of a B-scan image. It is available on OUP web in PC version (.exe) only.

b B-scan

i Production of a B-scan image

In B-scan, a fan beam of ultrasound pulses is used. The reflected pulses are detected and processed by a computer. A two-dimensional image is then produced.

The production of a B-scan image is illustrated in Figure 2.3g.

1 A transducer with an array of crystals emits a fan beam of ultrasound pulses into the body.

2 The transducer detects the echoes from interfaces along the path of each pulse.

This is the speed of ultrasound in soft tissue. ▶

3 The distance of the interface corresponding to each echo is calculated by assuming the speed of ultrasound is constant, usually 1540 m s^{-1}.

4 A trace with bright dots is produced for each path. The **position** of a dot on the trace indicates the **position of interface** and the **brightness** of the dot indicates the **intensity of the echo**.

✎ This method converts a one-dimensional line (A-scan) into a two-dimensional image (B-scan).

5 The traces are joined on a 2D plane to form a 2D image of the scanned area.

The brightness of a dot indicates the intensity of an echo. ▶

Note that the 'B' in **B**-scan stands for **brightness**.

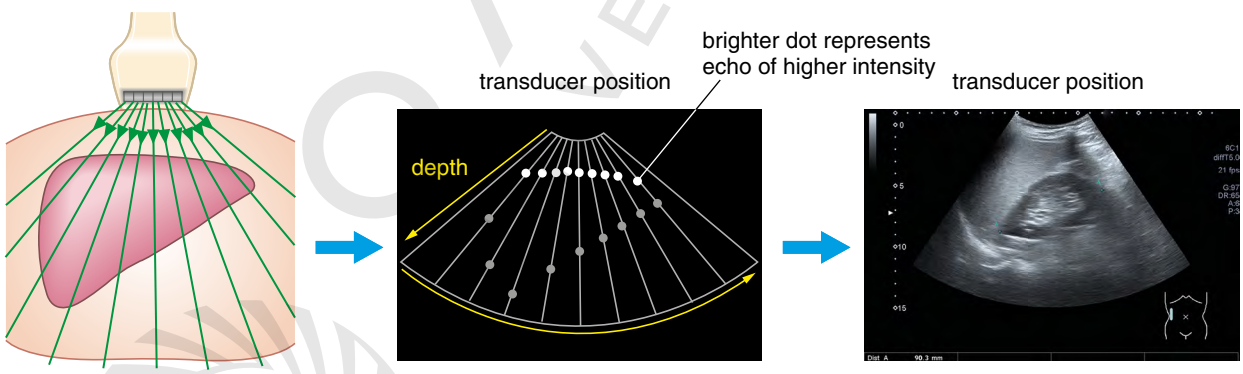


Fig 2.3g *Production of a B-scan image.*

The resulting image is a cross-section through the body scanned by the transducer. The doctor may sweep the transducer through the skin to view different cross-sections of the body in real time.

ii Interpreting a B-scan image

In order to observe the correct part and orientation of a body structure, skills are needed. The video below gives an example: ▶

Video

Interpreting a B-scan image is fairly straightforward as it is a cross-sectional image of the body in a suitable proportion (Fig 2.3h on p.94). Some tools are also provided in the computer software for further analysis.

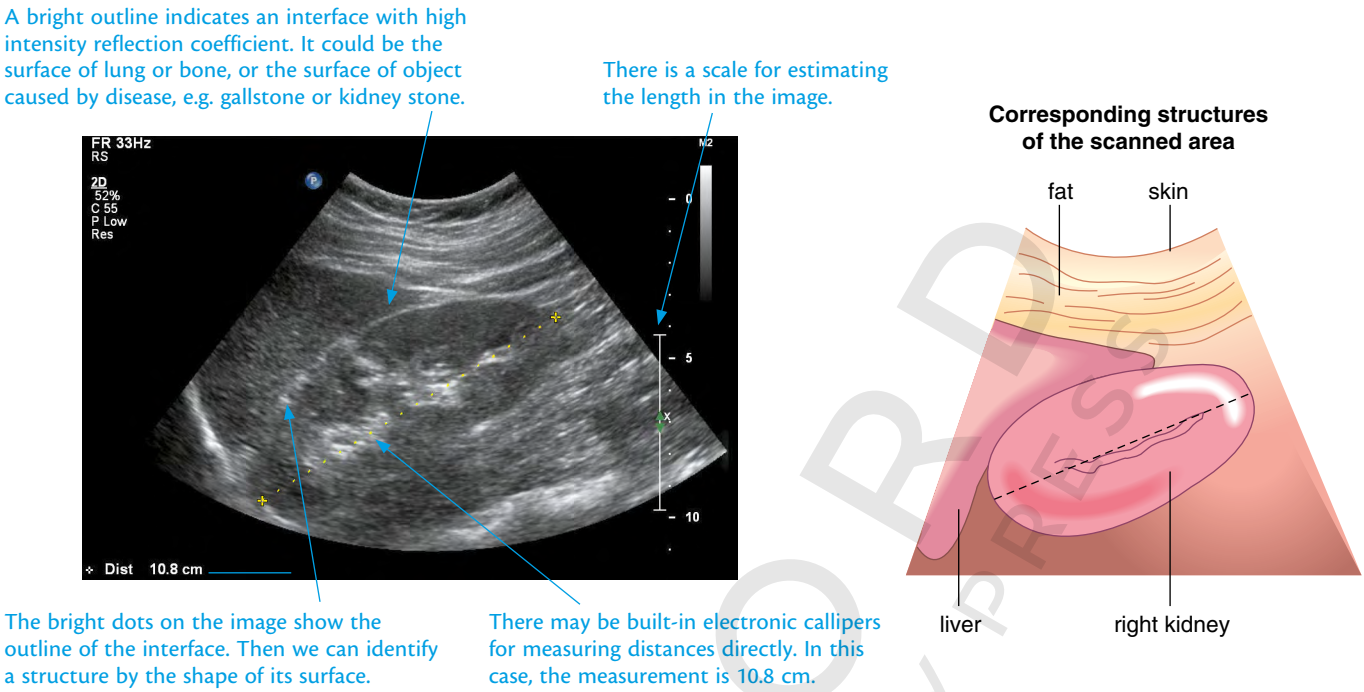


Fig 2.3h *Interpreting a B-scan image.*

Figure 2.3i shows an example of its application.

How can the size of the gallstone be estimated by using the side scale?

Solution

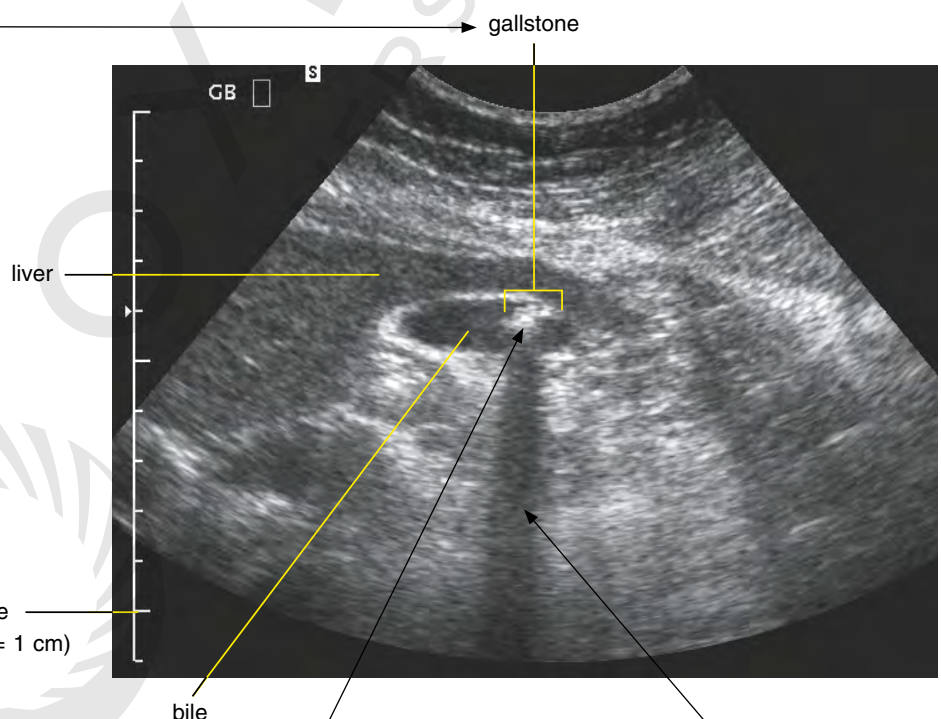
By directly measuring the lengths with a ruler, length of the 11 division side scale = 7.3 cm

length of the gallstone in the image = 0.8 cm

$\therefore$ Actual length of the gallstone

$$= 0.8 \times \frac{11}{7.3}$$

$$= 1.2 \text{ cm}$$



The gallstone appears to be bright while the bile in the gall bladder appears to be dark. What can you say about the reflection of ultrasound in the gall bladder?

Solution

Strong reflection occurs at the surface of the gallstone, while there is no significant reflection in the bile.

Why is there a dark band?

Solution

The surface of the gallstone reflects most of the ultrasound, thus echoes produced from structures behind it are very weak.

Fig 2.3i *An application of B-scan.*

Writing practice

Question

Explain why the surface of a gallstone appears to be bright in an ultrasound image. (2 marks)



Answer

The differences in acoustic impedances of the gallstone and the soft tissues around it is great, 1A

giving a large intensity reflection coefficient of the gallstone's surface. 1A

Therefore, the gallstone's surface appears to be bright.

Practise yourself

1 Is the bone surface easily distinguished in an ultrasound image? Explain your answer. (4 marks)

Google Forms Microsoft Forms

Quick check 12

→ E-version in OUP Exercise Platform is also available.

1 The brightness of dots on a B-scan image is related to the (intensity / time of detection) of echoes from the corresponding positions.

2 If we use an ultrasound B-scan system to observe the lungs, the interior of the lungs cannot be seen clearly because

A the lungs absorb almost all the ultrasound.

B the lungs absorb almost no ultrasound.

C almost all the ultrasound is reflected at the surface of the lungs.

D almost no ultrasound is reflected at the surface of the lungs.

DSE goal

Realize dependence of attenuation of ultrasound on nature of medium and frequency.

Identify suitable frequency ranges of ultrasound for scanning.

DSE exam

16(2)Q4.6, 18(2)Q4.6(b), 22(2)Q4.6, 23(2)Q4.3

5 Choosing suitable frequencies for ultrasound scanning

There are several factors affecting the ultrasound frequencies we use for medical scanning.

a Penetration depth

When an ultrasound pulse travels through a medium, its intensity is gradually reduced as some ultrasound is scattered or absorbed by the particles in the medium (Fig 2.3j). This reduction in intensity is called **attenuation**.

Scattering refers to the deflection of a wave by the particles in the medium through which it passes.

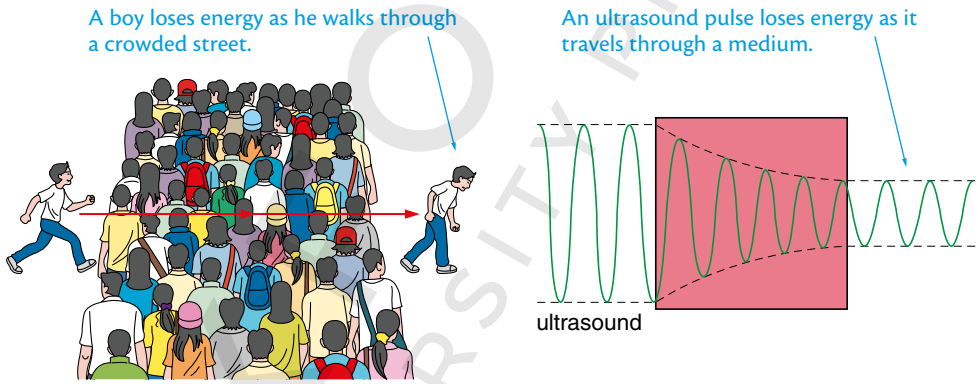


Fig 2.3j *Analogy of attenuation of ultrasound. In this analogy, a boy walking through a crowded street represents an ultrasound pulse travelling through a medium.*

The ultrasound frequency can affect the degree of attenuation and hence the penetration depth (penetrating power). As the ultrasound frequency increases, the pulse attenuates more and therefore its penetration depth decreases.

$$f \uparrow \Rightarrow \textbf{attenuation} \uparrow \Rightarrow \textbf{penetration depth} \downarrow$$

The degree of attenuation also depends on the medium through which ultrasound travels. Figure 2.3k shows a rough comparison of attenuation in some common body tissues.

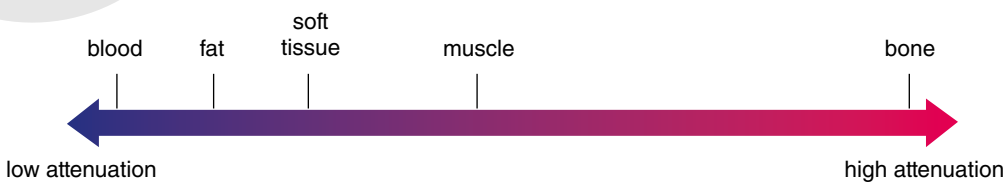


Fig 2.3k *Rough comparison of attenuation in different body tissues.*

attenuation 衰減　　scattering 散射

b Axial resolution

In ultrasound scanning, as each beam of ultrasound pulse travels along its own path, it is important to see **whether the echoes from two close interfaces along the path can be identified**.

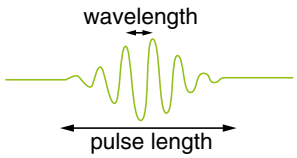


Fig 2.3l Wavelength and pulse length.

An ultrasound with a higher frequency has a shorter wavelength. A shorter wavelength allows the ultrasound to have a shorter pulse length (Fig 2.3l), making the echoes from two close interfaces less likely to overlap (Fig 2.3m). As a result, smaller details along the axial direction (along the beam direction) can be displayed on the ultrasound image. We say that the image has a higher **axial resolution**.

$$f \uparrow \Rightarrow \textbf{axial resolution} \uparrow$$

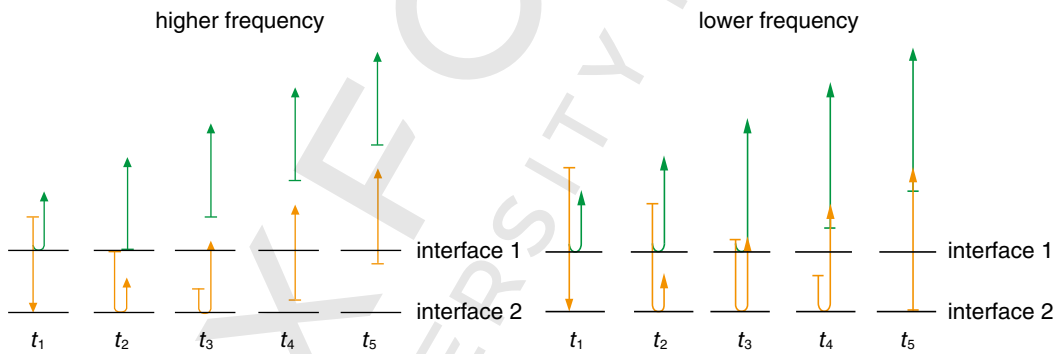


Fig 2.3m Echoes from two interfaces along the same path.

c Typical frequencies for different body structures

To choose suitable frequencies for ultrasound scanning, you should determine whether a larger penetration depth or a higher resolution is needed. You can never get both. Table 2.3a shows some ultrasound frequencies used in ultrasound scanning.

Frequency range / MHz	Properties	Examples of body part
10–12	Resolution: ☺☺☺ Penetration depth: ☺	neck, breasts, thyroid, testicles
7–8	Resolution: ☺☺ Penetration depth: ☺☺	ovaries, uterus, prostate
3–5	Resolution: ☺ Penetration depth: ☺☺☺	abdominal organs, e.g. liver, kidneys

small and near the skin surface

large and deep inside the body

Table 2.3a Some typical ultrasound frequencies used in ultrasound scanning.

axial resolution 軸向分辨率

Know more

Hearing through the wall

You may hear your neighbours talking on the other side of a wall but it is hard to work out the details of the conversation. Why?

Higher frequency sounds are mostly attenuated by the wall, so only the lower frequency sounds reach your ear. This is why a normal conversation on the other side becomes mumblings on your side.



Google Forms Microsoft Forms

Quick check 13

→ E-version in OUP Exercise Platform is also available.

1 *True or false*: As an ultrasound pulse travels through the human body, it loses energy. (T/F)

2 The higher the ultrasound frequency, the (higher / lower) the attenuation and the (longer / shorter) the penetration depth.

3 Ultrasound with high frequency has (high / low) penetrating power and produces images with (high / low) resolution. It is suitable for imaging body parts (deep inside the body / near the skin surface).

4 Which of the following body parts would need the lowest frequency of ultrasound when performing a scan?

A Breast B Eye C Neck D Liver

DSE goal

Discuss advantages and limitations of ultrasound scans in diagnosis.

DSE exam

12(2)Q.4(b)(ii), 13(2)Q4.7, 14(2)Q.4(a)(ii)

Cross-link

Imaging methods involving ionizing radiation will be discussed in Ch 3. ▶

6 Advantages and limitations of ultrasound scanning in diagnosis

Using ultrasound scanning in diagnosis has several advantages and limitations (Table 2.3b).

Advantages
• It **does not involve ionizing radiation**.
• It can detect the movement of an organ in **real time**, e.g. heart, diaphragm.
• It is **readily accessible**. The equipment is portable and convenient to use.
• **No discomfort** is caused to patient.
Limitations
• It has **limited tissue penetration**, particularly through bones or air-filled organs such as the lung and the *bowel*.
• Its **field of view is relatively small**.

In comparison, inserting an endoscope into a patient's body will cause some degree of discomfort. ▶

Most ultrasound is reflected at the surfaces of bones or air-filled organs. ▶

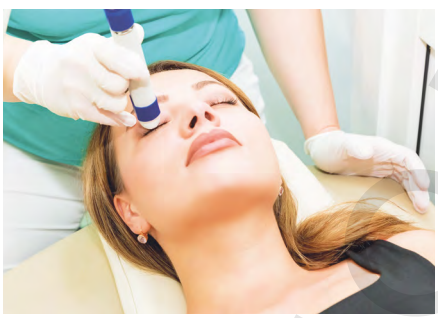
Table 2.3b *Advantages and limitations of ultrasound scanning.*

bowel 腸臟

7 Common uses of A-scan and B-scan

Both ultrasound A-scan and B-scan have a wide range of applications.

Recall that A-scan produces a one-dimensional image (only depth is shown). It is typically used to study the eye (Fig 2.3n), such as measuring the lens-to-retina distance. It is also used to identify different layers in complex structures such as the brain.



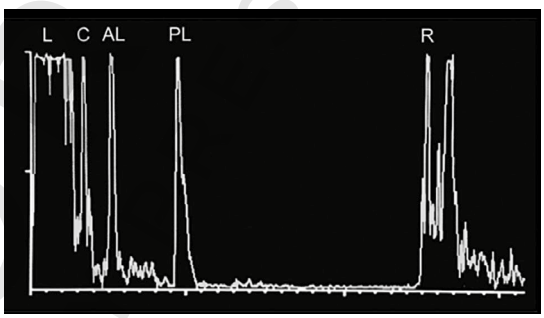


Fig 2.3n *A-scan for eye.*

On the other hand, B-scan produces a two-dimensional image (depth and width are shown). A typical use of B-scan is to monitor foetuses during pregnancy (Fig 2.3o). You probably had one before you were born. B-scan is also used to produce images of other body parts such as muscles, various glands on the neck and organs near the pelvis.



Fig 2.3o *B-scan for foetus.*

Know more

Other medical uses of ultrasound

Besides imaging, is there any other medical use of ultrasound?

▸ p.114

Exercise 2.3

★**1** Which of the following is an advantage of using ultrasound with higher frequency in medical imaging?

A It causes less pain to patient.

B Its penetrating power is higher.

C It improves the axial resolution.

D It carries less energy and is safer to use.

★**2** In a B-scan image, certain positions appear bright because

A no ultrasound attenuation occurs there.

B no ultrasound reflection occurs there.

C strong ultrasound attenuation occurs there.

D strong ultrasound reflection occurs there.

★**3** Which of the following statements about A-scan and B-scan are correct?

(1) Both allow us to see the shape of an internal organ.

(2) Both show the intensity of the ultrasound reflected.

(3) Both can be used to measure the depth of body tissue.

A (1) and (2) only B (1) and (3) only

C (2) and (3) only D (1), (2) and (3)

4 When an ultrasound pulse travels through a medium, attenuation occurs.

(a) What is attenuation?

(b) Give two factors which affect the attenuation of the pulse.

5 Suggest one advantage and one disadvantage of ultrasound B-scan when compared to optical fibre endoscopy.

6 State a major difference between an A-scan image and a B-scan image.

7 It is known that the axial resolution in ultrasound imaging is inversely proportional to the frequency of ultrasound used. If the axial resolution is 1.1×10^{-4} m when using ultrasound at a frequency of 7 MHz, what is the axial resolution when using 10 MHz?

8 An ultrasound image of a foetus is shown in Figure a.

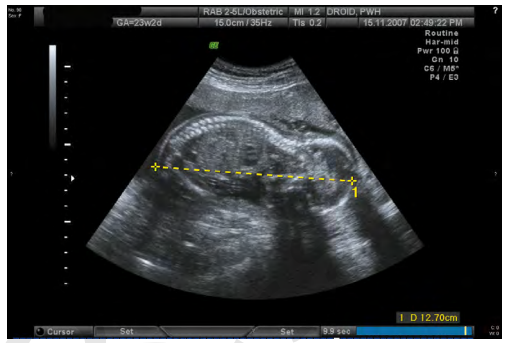


Fig a

(a) Explain why the spine of the foetus appears to be bright in the image.

(b) Give one reason to explain why ultrasound is suitable for imaging a foetus.

★**9** (a) How should the frequency of ultrasound be changed in order to distinguish soft tissues with a smaller axial separation during ultrasound scanning?

(b) State and explain a consideration regarding the choice of frequency for ultrasound imaging other than axial resolution.

★**10** A transducer sends an ultrasound pulse towards a patient's liver. The following image is produced and displayed on the monitor (Fig b).

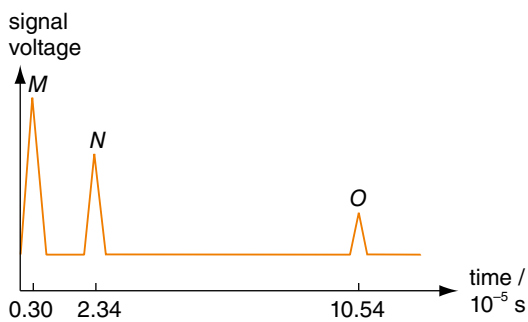


Fig b

(a) Is it an A-scan image or a B-scan image?

(b) Spike *M* is not produced by the echo from the liver's surface. Suggest how it is produced.

(c) Spikes *N* and *O* are produced by the echoes from the liver's front and back surfaces. Estimate the distance across the liver along the propagating direction of the pulse. Take the speed of ultrasound in liver be 1540 m s^{-1}.

*11 Figure c shows an ultrasound transducer placed on the skin. The A-scan display is shown in Figure d.

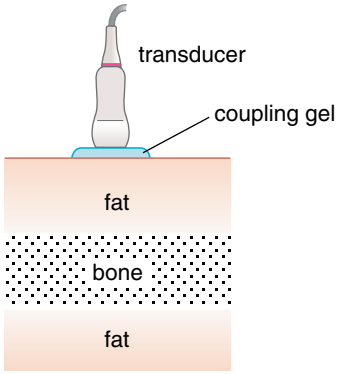


Fig c

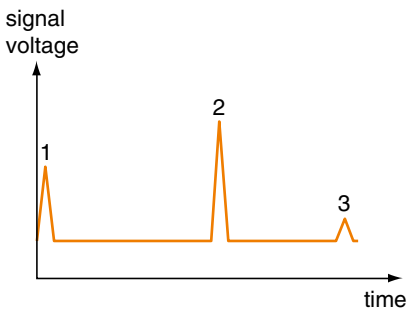


Fig d

(a) What does spike 2 denote?

(b) Suggest a possible reason to explain why spike 2 is taller than

(i) spike 1;

(ii) spike 3.

*12 Performing a kidney scan usually needs ultrasound with lower frequency than performing a thyroid scan. Explain the reason.

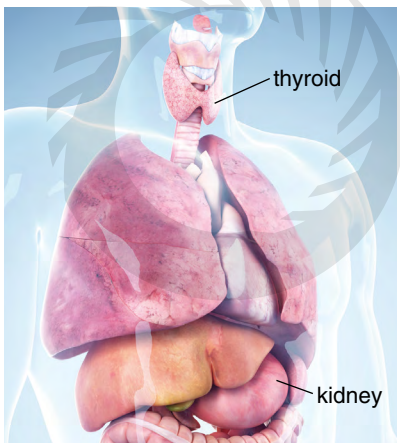


Fig e

*13 Figure f shows a B-scan image of a 23-week foetus.

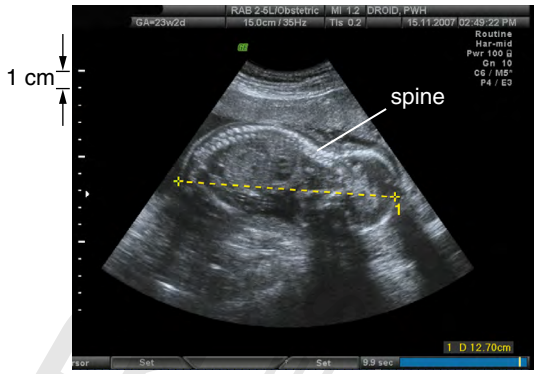


Fig f

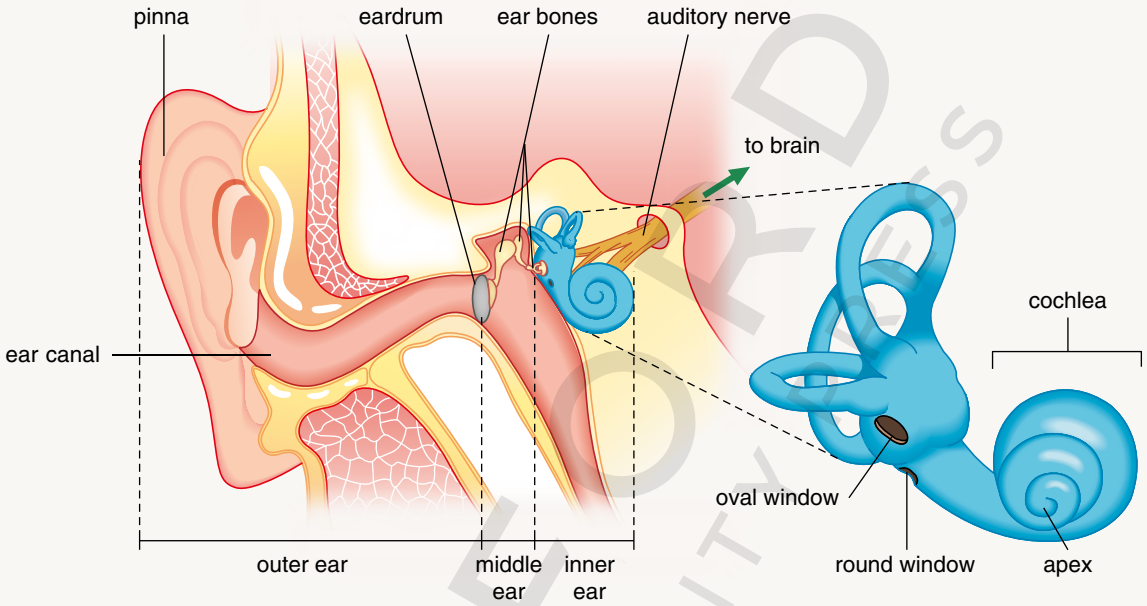
(a) Using the spine of the foetus as an example, describe how the bright part of the image is produced.

(b) Given 1 mark in the side scale represents 1 cm, estimate the body length of the foetus as indicated by the callipers (yellow crosses).

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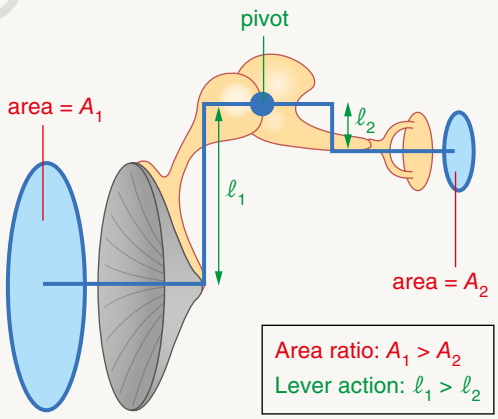
Key concepts 2

1 Structure of the human ear



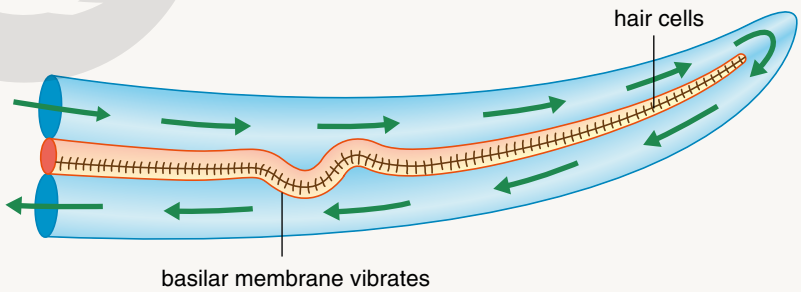
2 Pressure amplification

The middle ear amplifies pressure by lever action of the ear bones and the area ratio of the eardrum and the oval window.



3 Converting mechanical vibrations into electrical signals

The basilar membrane is set into vibration by sound waves entering the cochlea. Hair cells are stimulated and send nerve pulses to the brain.



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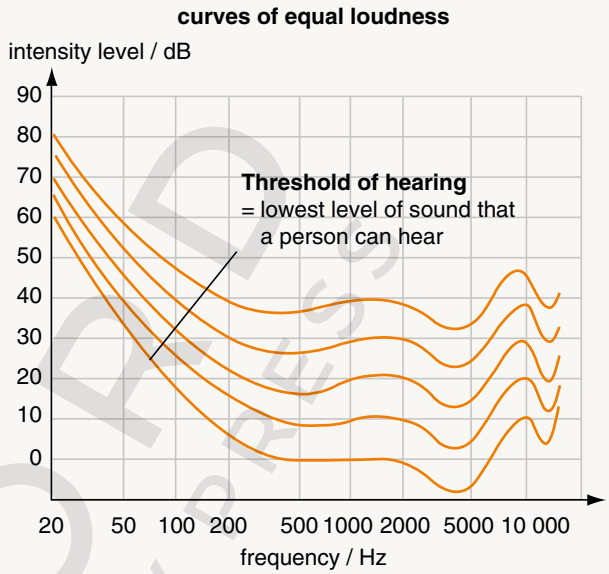
4 Perceived loudness and hearing loss

Sound intensity level

- Unit: decibel (dB)
- $L = 10 \log \left(\frac{I}{I_0} \right)$
- $I_0 = 10^{-12}$ W m^{-2}

Perceived loudness

- Unit: phon
- A sound of n phons has the same perceived loudness as a 1000-Hz pure note at n dB

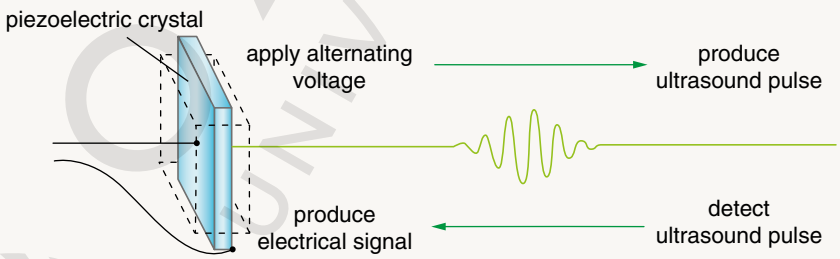


Hearing loss

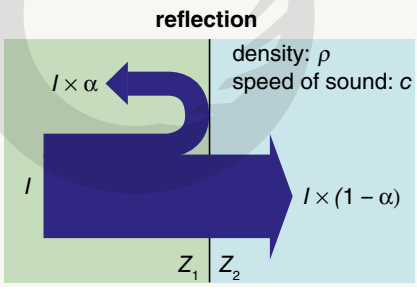
can be resulted from repeated exposure to loud noise or a single exposure to noise above 140 dB

5 Production and detection of ultrasound

An ultrasound transducer produces and detects ultrasound by piezoelectric crystals.



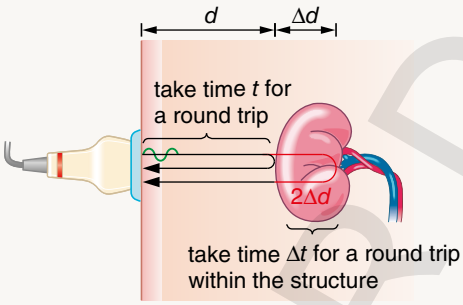
6 Reflection of ultrasound



- acoustic impedance $Z = \rho c$
- intensity reflection coefficient $\alpha = \frac{(Z_2 - Z_1)^2}{(Z_2 + Z_1)^2}$

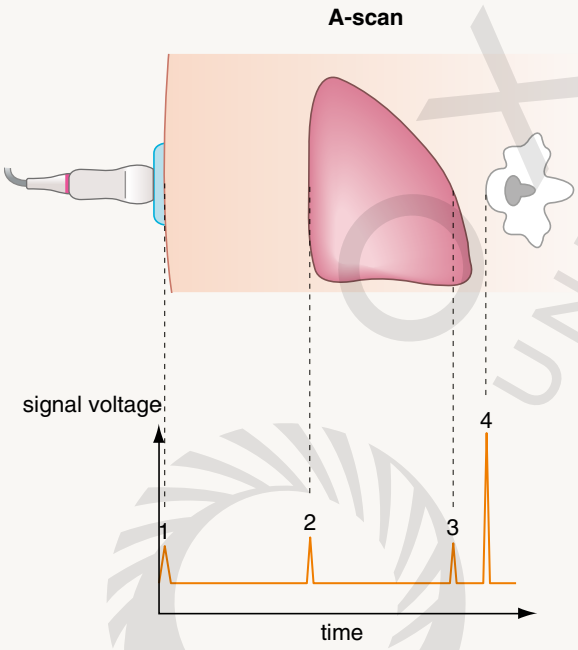
7 Range measurement by ultrasound

The distance of an interface from the transducer can be calculated from the time of receiving an echo. Take the speed of the ultrasound pulse as c.

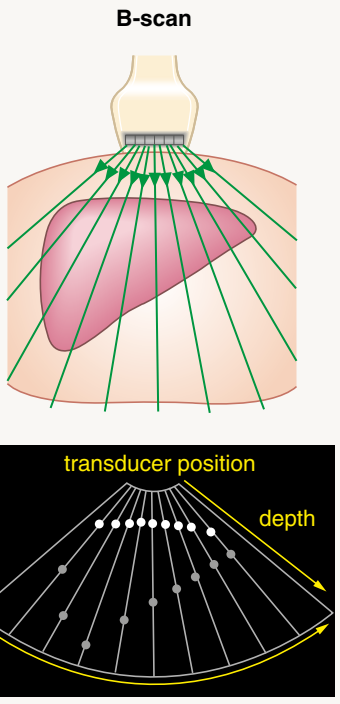


$$d = \frac{1}{2}ct \quad \text{and} \quad \Delta d = \frac{1}{2}c\Delta t$$

8 A-scan and B-scan

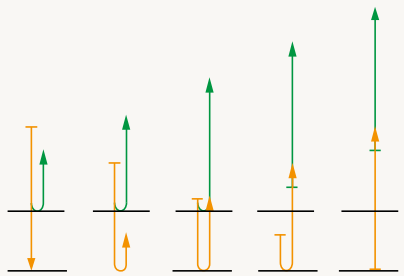
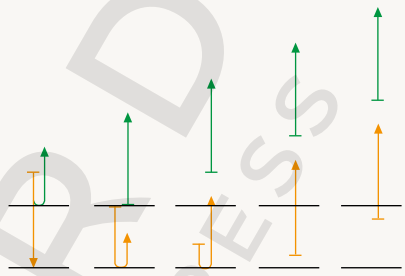


- transducer output connected to oscilloscope
- reflected pulse from each interface is represented by a spike
- x-axis represents time of detection
- amplitude of spike represents pulse strength



- fan beam of ultrasound pulses is emitted to the body
- reflected pulse from each interface is represented by a bright dot
- brightness of dot represents pulse strength

9 Choosing suitable frequencies for ultrasound scanning

lower frequency	higher frequency
lower axial resolution 	higher axial resolution 
larger penetration depth	smaller penetration depth
used for imaging structures which are large and deep inside the body	used for imaging structures which are small and near the skin surface

Q20 Exam Report: Unsatisfactory. In (a)(i), many candidates failed to use the graph to get the time that ultrasound travelled within each layer. Some candidates mixed up the time of *AC* with the time of individual layers. In (a)(ii), some candidates wrongly applied the acoustic impedance of soft tissue, which is in fact redundant. Quite a lot of them gave a wrong unit for density (e.g. kg m^{-1}, kg m^{-2}, kg m^{3}). The performance in (b)(i) was poor. Many candidates did not state clearly the location where reflection occurs. Not many explained how a 2-D image is produced. Part (b)(ii) was well answered. Some candidates thought that radiation is equivalent to ionizing radiation and did not realise that ultrasound is in fact one kind of radiation.

Q21 Exam Report: Part (a)(i) was well answered. Many failed to deduce the gain of pressure in (a)(ii). In (b)(i), some candidates were unable to state concisely the physical significance of the loudness curve being higher at both low and high frequencies. Not many candidates managed to identify the correct loudness curve in (b)(ii) with appropriate explanations. Part (c) was in general well answered although a few of them were not familiar with logarithmic operations.

Key terms

1 acoustic impedance 聲阻抗 p.80
2 A-scan A-掃描 p.89
3 attenuation 衰減 p.96
4 auditory nerve 聽神經 p.63
5 axial resolution 軸向分辨率 p.97
6 basilar membrane 基底膜 p.63
7 B-scan B-掃描 p.89
8 cochlea 耳蝸 p.61
9 coupling gel 耦合凝膠 p.84
10 curve of equal loudness 等響曲線 p.70
11 decibel (dB) 分貝 p.67
12 ear bone / ossicle 聽小骨 p.62
13 eardrum 鼓膜 p.62
14 hair cell 毛細胞 p.63
15 inner ear 內耳 p.61
16 intensity reflection coefficient 反射聲強係數 p.82
17 middle ear 中耳 p.61
18 outer ear 外耳 p.61
19 oval window 卵圓窗 p.62
20 phon 方 p.70
21 piezoelectric effect 壓電效應 p.77
22 sound intensity level 聲強級 p.67
23 threshold of hearing 聽覺閾 p.67
24 ultrasound scanning 超聲波掃描 p.76

Revision 2

Multiple-choice questions

★1 Misc Which of the following is/are reason(s) why the pressure applied on the oval window is larger than the pressure applied on the eardrum?

(1) Lever action occurs at the middle ear.

(2) The oval window has a smaller area than the eardrum.

(3) The acoustic impedance in the inner ear is much higher than that in the outer ear.

A (1) and (2) only

B (1) and (3) only

C (2) and (3) only

D (1), (2) and (3)

★2 2.1 Darwin is operating a machine which produces sound. The intensity of the sound is 3×10^{-3} W m^{-2} when reaching his ears. By wearing ear protectors, the sound intensity received by Darwin drops to 1×10^{-6} W m^{-2}. Find the reduction in the sound intensity level.

A 4.8 dB B 34.8 dB

C 60 dB D 94.7 dB

★3 2.2 A sound wave in a medium with acoustic impedance Z_1 enters normally into another medium with acoustic impedance Z_2. If the intensity reflection coefficient is 0.25, which of the following is a possible ratio of Z_1 to Z_2?

A 1 : 1 B 2 : 3

C 3 : 1 D 5 : 3

★4 2.3 The figure below shows an A-scan display of the spleen (Fig a). Spikes *X* and *Y* indicate the echoes from the front surface and the back surface of the spleen respectively.

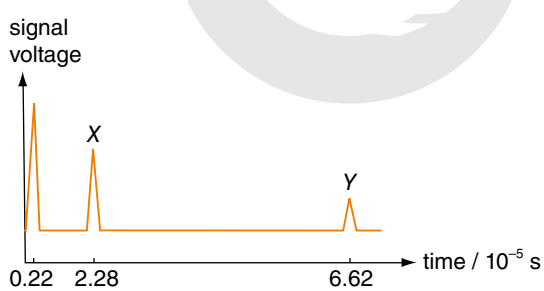


Fig a

What is the depth of the back surface of the spleen under the skin? Take the speed of ultrasound along its path to be 1565 m s^{-1}.

A 1.61 cm B 3.40 cm

C 5.01 cm D 5.18 cm

★5 2.1 Figure b shows several curves of equal loudness of a person.

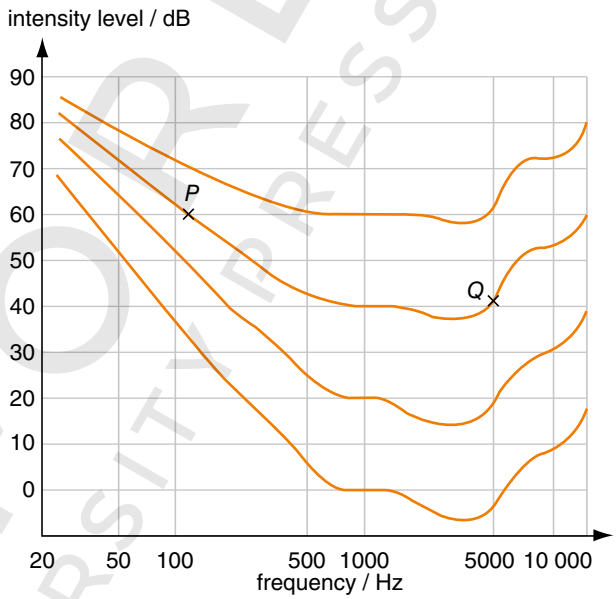


Fig b

Which of the following statements is/are correct?

(1) The lowest curve indicates the threshold of hearing of the person.

(2) Note *P* is louder to the person than note *Q*.

(3) The intensity of note *P* is about 100 times higher than that of *Q*.

A (1) only

B (1) and (3) only

C (2) and (3) only

D (1), (2) and (3)

★6 2.1 Karl is at a certain distance *d* from a point source of sound. How should he change his distance from the sound source to reduce the intensity level of sound reaching him by 20 dB?

A Change to 2*d*

B Change to 10*d*

C Change to 20*d*

D Change to 100*d*

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**7 ★★ 2.3 An ultrasound pulse travels through a layer of fat with thickness Δd_1 and a bone with thickness Δd_2, as shown in Figure c. Figure d shows the corresponding A-scan image produced.

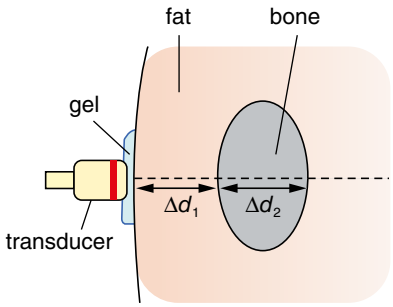


Fig c

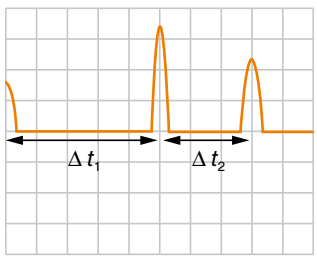


Fig d

The speeds of sound in the fat and the bone are respectively c_1 and c_2 ($c_1 \neq c_2$).

Which of the following is equal to $\frac{c_1}{c_2}$?

A $\frac{\Delta d_1}{\Delta d_2}$ B $\frac{\Delta t_2}{\Delta t_1}$

C $\frac{\Delta d_1 \Delta t_1}{\Delta d_2 \Delta t_2}$ D $\frac{\Delta d_1 \Delta t_2}{\Delta d_2 \Delta t_1}$

Refer p.86, 87

8 HKDSE 2013 Paper 2 Q4.5 2.1

A speaker is connected to an amplifier to produce sound. When the power supplied to the speaker is 50 W, the resulting sound intensity level at a certain location is 100 dB. Assume that there is no other sound source and the speaker has a fixed efficiency of converting electrical energy to sound. What is the power required to produce a sound intensity level of 110 dB at the same location?

A 52 W B 55 W

C 100 W D 500 W (37%)

9 HKDSE 2015 Paper 2 Q4.3 Misc

The following diagram shows an ultrasound scanner with an array of crystals. A pair of electrodes is connected across each crystal. Which of the following statements is/are correct?

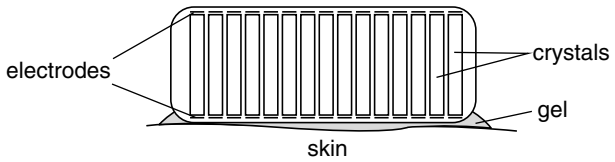


Fig e

(1) The gel is used to reduce the attenuation of ultrasound when passing the gap between the scanner and the skin.

(2) When a crystal receives ultrasound, an electrical signal is generated between the electrodes by piezoelectric effect.

(3) This scanner is designed for taking B-scan images.

A (1) only B (2) only

C (1) and (3) only D (2) and (3) only (26%)

10 HKDSE 2015 Paper 2 Q4.4 Misc

Ultrasound scanning is **not** suitable for lungs because air in the lungs

A has a high attenuation coefficient and can absorb almost all the ultrasound.

B has a low attenuation coefficient and can absorb almost all the ultrasound.

C has high acoustic impedance compared to soft tissue and can reflect almost all the ultrasound.

D has low acoustic impedance compared to soft tissue and can reflect almost all the sound. (29%)

11 HKDSE 2017 Paper 2 Q4.2 2.1

Which of the following statements about human hearing are correct?

(1) The ear bones in the middle ear convert sound waves into vibrations of the ear drum.

(2) Pressure is amplified because of the difference in area between the ear drum and the oval window.

(3) Mechanical vibrations are converted into electrical signals in the inner ear.

A (1) and (2) only

B (1) and (3) only

C (2) and (3) only (40%)

D (1), (2) and (3)

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12 **HKDSE 2019 Paper 2 Q4.3**
2.1

The maximum sensitivity of human ear to sound of frequency 3 kHz is about 0.5 dB, which is the minimum change in sound intensity level that can be detected by the ear. This corresponds to a change of sound intensity of approximately

A 12%. (40%)

B 6%.

C 3%.

D 1%.

13 **HKDSE 2020 Paper 2 Q4.4**
2.2

Which of the following statements about the piezoelectric crystal inside an ultrasound transducer is/are correct?

(1) The piezoelectric crystal converts electrical signals into mechanical vibrations and vice versa.

(2) The thickness of the piezoelectric crystal is arbitrary.

A Only (1) is correct. (46%)

B Only (2) is correct.

C (1) and (2) are correct.

D (1) and (2) are incorrect.

Conventional questions

14 Misc

Figure f shows an ultrasound image of an internal organ. The ultrasound pulse enters the body through the skin. It then travels from medium *X* to medium *Y*, and finally to medium *Z*. Assume the acoustic impedances of media *X* and *Z* are the same.

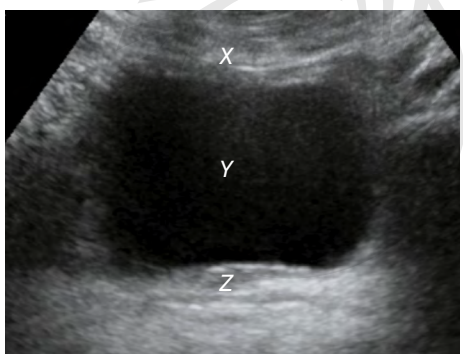


Fig f

(a) Johnson thinks that region *Y* appears dark because the attenuation of ultrasound in it is small. Comment on this statement. (2 marks)

(b) Suppose an ultrasound pulse of intensity 0.01 W m^{-2} is incident on the *X*–*Y* interface and the intensity reflected is 0.001 W m^{-2}. Find the intensity reflected at the *Y*–*Z* interface, assuming that the attenuation in region *Y* is negligible. (3 marks)

***15** Misc

(a) An ultrasound pulse travels from body fat to muscle. The acoustic impedances of body fat and muscle are 1.38×10^6 kg m^{-2} s^{-1} and 1.70×10^6 kg m^{-2} s^{-1} respectively.

(i) Find the intensity reflection coefficient between body fat and muscle. (2 marks)

(ii) The intensity of the ultrasound pulse incident on the fat-muscle interface is 40 W m^{-2}. Find the intensity of the reflected pulse. (2 marks)

(iii) When the reflected pulse is received by the transducer, its intensity is detected to be 0.237 W m^{-2}. Suggest a reason for difference between this value and the answer in (a)(ii). (1 mark)

(b) In an ultrasound scan, an ultrasound pulse is sent by an ultrasound transducer towards the organ in the body to be examined. Explain how the distance of the organ from the transducer can be computed. Assume the ultrasound pulse travels in the body at a constant speed *c*. (2 marks)

***16** Misc

(a) (i) Describe how an ultrasound transducer produces an ultrasound pulse. (2 marks)

(ii) State the energy conversion in the transducer when it produces an ultrasound pulse. (1 mark)

(b) Ronald is having an ultrasound examination of his eye. During the examination, a transducer is placed on his eye (Fig g). The echoes from interfaces *A*, *B*, *C* and *D* are recorded in the A-scan display (Fig h).

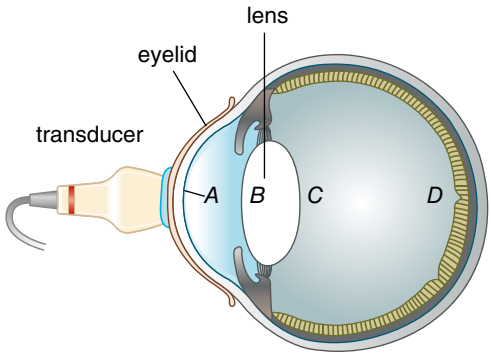


Fig g

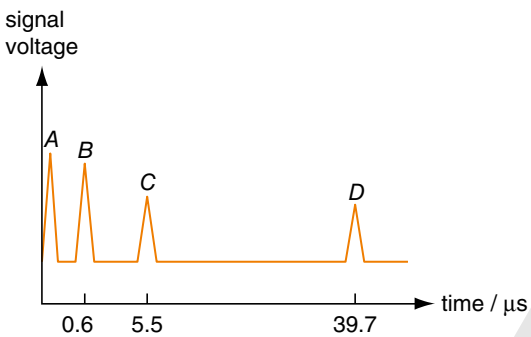


Fig h

Take the speed of ultrasound in eye investigations to be 1500 m s^{-1}.

(i) Find the thickness of the lens. (2 marks)

(ii) The acoustic impedance of the lens is 1.84×10^6 kg m^{-2} s^{-1}. Find the density of the lens. (2 marks)

★17 Misc A patient is having an ultrasound examination. An ultrasound pulse from the transducer enters his body and encounters interfaces *P*, *Q*, *R* and *S* (Fig i). Figure j shows the resulting A-scan image.

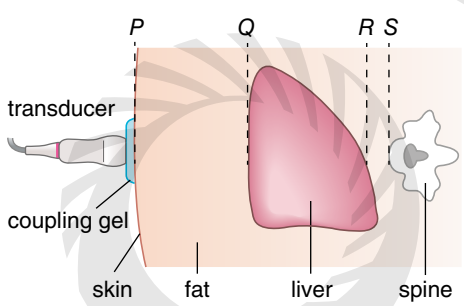


Fig i

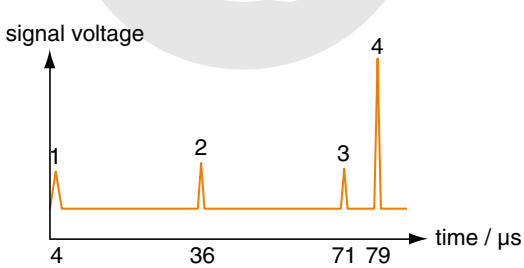


Fig j

(a) Which spikes correspond to the ultrasound pulses reflected from interfaces *P* and *S*? (2 marks)

(b) What is the depth of the spine under the skin? Take the speed of sound in soft tissue and fat to be 1540 m s^{-1}. (2 marks)

(c) (i) Explain why spike 4 is taller than spike 3. (2 marks)

(ii) Explain why spike 3 is shorter than spike 2. (2 marks)

★18 2.1 Figure k shows some curves of equal loudness of Janet.

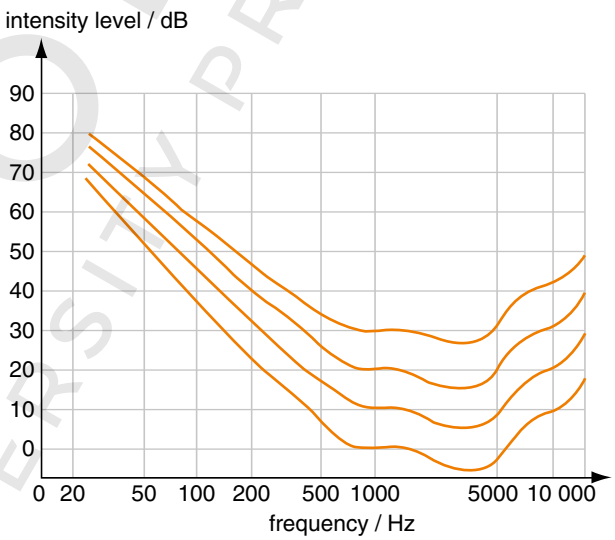


Fig k

(a) What does a curve of equal loudness indicate? (1 mark)

(b) Janet hears a 10-kHz note at 20 dB.

(i) Estimate the phon number of this note. (1 mark)

(ii) If the frequency of the note decreases gradually from 10 kHz to 10 Hz while the sound intensity level remains unchanged, how will the loudness of the note change as perceived by Janet? (2 marks)

(c) Janet is exposed to a 500 Hz note with an intensity of 3.5×10^{-9} W m^{-2}. Can this note be heard by Janet? Explain your answer with calculation. Take $I_0 = 10^{-12}$ W m^{-2}. (4 marks)

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**19 Misc

Karen is having an ultrasound A-scan imaging. The doctor applies a layer of coupling gel between the transducer and the skin (Fig l). The transducer sends an ultrasound pulse at time $t = 0$ and receives the echo from interface A after 8 μs (Fig m).

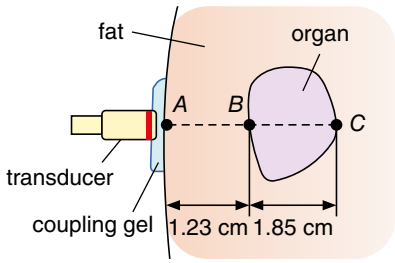


Fig l

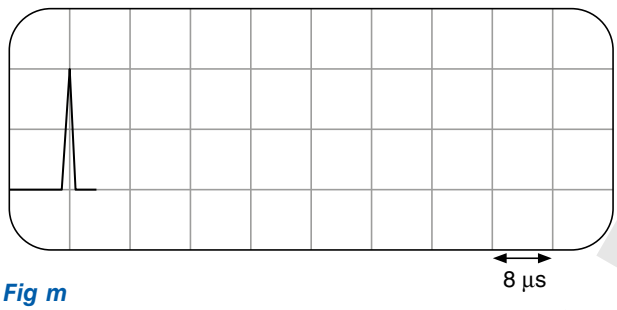


Fig m

Take the speed of ultrasound in fat and organ to be 1540 m s^{-1}.

(a) Estimate the time taken for the pulse to travel

(i) from interface A to interface B;

(ii) from interface B to interface C. (3 marks)

(b) Draw the spikes displayed on Figure m corresponding to interfaces B and C. Assume that all the echoes received by the transducer have the same intensity. (2 marks)

(c) How will the A-scan image be affected if no coupling gel is applied? (1 mark)

(d) The density of the organ is 1060 kg m^{-3}.

(i) Find the acoustic impedance of the organ.

(ii) If the acoustic impedance of fat is 1.38×10^6 kg m^{-2} s^{-1}, how many percentage of the intensity of the ultrasound pulse will be reflected at interface B? (4 marks)

▶ Refer p.83, 89

See p.105 for the Exam Report.

**20 Misc

HKDSE 2012 Paper 2 Q.4

(a) Figure n shows a cross-section of a piece of bone with 5.8 cm thickness situated below a layer of soft tissue that is 2.0 cm thick. An ultrasound transducer with coupling gel is applied to the skin. The ultrasound pulses reflected from various interfaces A, B and C are displayed on a CRO.

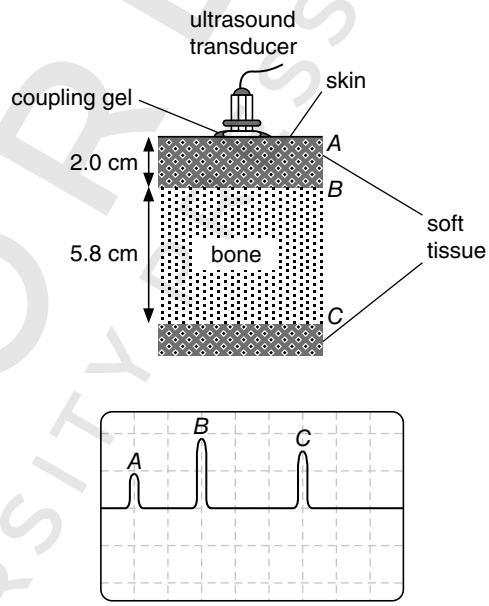


Fig n

(i) Find the ratio of the speed of ultrasound in bone to that in soft tissue. (2 marks)

The values of acoustic impedance of various body tissues to the ultrasound used are tabulated below.

Tissues	Acoustic impedance / kg m^{-2} s^{-1}
soft tissue (average)	1.63×10^6
bone	7.78×10^6

Table a

(ii) If the speed of ultrasound in soft tissue is 1580 m s^{-1}, estimate the density of bone. (3 marks)

(b) (i) Describe the working principles of ultrasound B-scan imaging. (3 marks)

(ii) State one advantage and one limitation of using ultrasound scans in the context of medical imaging. (2 marks)

See p.105 for the Exam Report.

21 HKDSE 2015 Paper 2 Q.4

2.1

Figure o shows the structure of human ear.

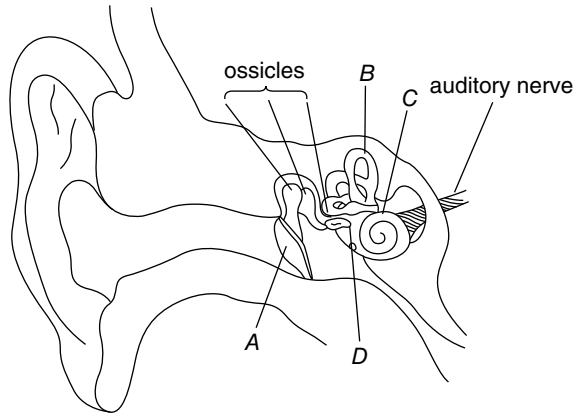


Fig o

(a) (i) Match the letters *A*, *B*, *C* and *D* in Figure o with the parts of the ear, namely, **eardrum**, **oval window**, **semi-circular canals** and **cochlea**. State the function of *C*. (2 marks)

(ii) The ratio of area of *A* to that of *D* is 20. If the ear amplifies the pressure of a sound signal by 25 times totally after passing *D*, find the gain of pressure contributed by the lever action of the ossicles. (1 mark)

(b) Figure p shows an equal loudness curve of people with normal hearing.

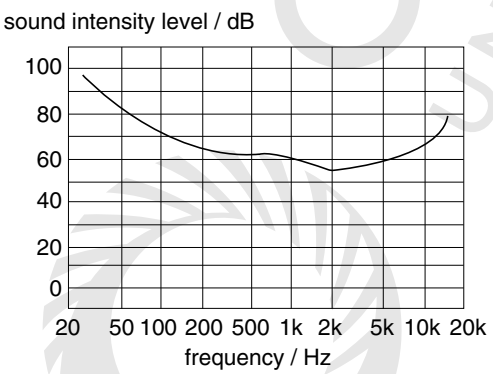


Fig p

(i) State the loudness, in phons, represented by this curve. State the physical significance of the curve being higher up at both ends. (2 marks)

(ii) A worker has suffered from hearing loss due to prolonged exposure to a noisy environment and the loss is most severe for sounds around kHz frequencies. If the worker is tested for the threshold of hearing, which of the equal loudness curves, *A*, *B* and *C*, in Figure q best represents his response? Explain your choice. (2 marks)

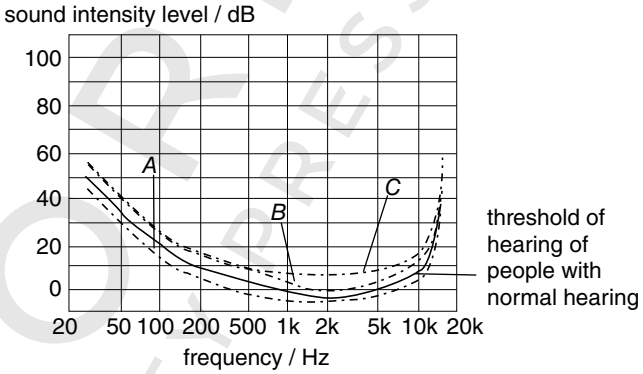


Fig q

(c) An engineer working near an operating engine is exposed to noise of sound intensity 80 W m^{-2}. After putting on earmuffs, the sound intensity of the noise heard is reduced to 2.5×10^{-5} W m^{-2}. Estimate the decrease in sound intensity level of the noise, in dB, heard after wearing the earmuffs. (3 marks)

See p.113 for the Exam Report.

22 HKDSE 2018 Paper 2 Q.4

Misc

(a) Piezoelectric transducers can generate as well as detect ultrasound. Figure r shows the basic construction of a piezoelectric transducer in which the thickness of the piezoelectric crystal is *t*.

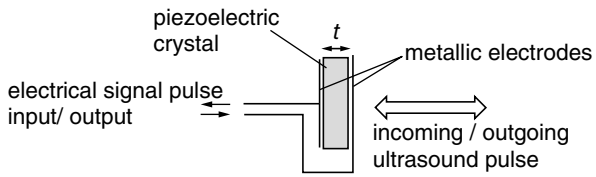


Fig r

(i) The crystal has a maximum response for a certain frequency *f* of the electrical signal applied such that *t* is about half of the wavelength of the ultrasound in the crystal. The speed of ultrasound in the crystal is 4000 m s^{-1}. Estimate *f* if *t* = 0.4 mm. (1 mark)

(ii) An ultrasound pulse produced by the piezoelectric transducer is reflected back from an interface as an echo. Explain how the transducer detects this echo. (2 marks)

(b) For medical imaging using an ultrasound beam, the resolution (in mm) along the beam direction (axial direction) in soft tissue varies with the frequency (in MHz) of the ultrasound used as shown below.

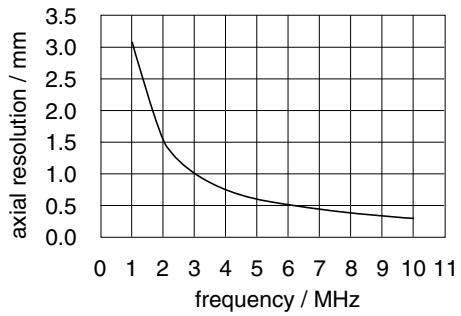


Fig s

(i) Explain the importance of axial resolution in ultrasound imaging. (2 marks)

(ii) In fact the axial resolution is inversely proportional to the frequency of ultrasound used. Make use of one or two points of the graph to deduce the axial resolution of imaging when using 12 MHz ultrasound. (2 marks)

(c) Figure t shows the set-up for studying a tissue interface underneath a 2 cm thick fat layer and a 8 cm thick of soft tissue layer. The speed of ultrasound in the fat layer is 1450 m s^{-1} while that in the soft tissue layer is 1540 m s^{-1}.

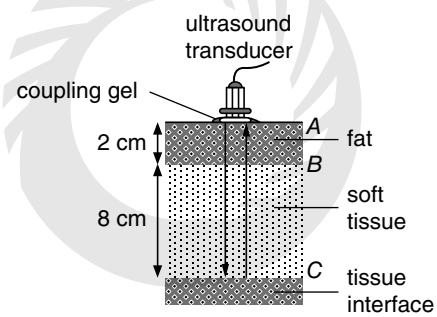


Fig t

(i) Calculate the echo time T for the ultrasound going to and from the tissue interface at C. (2 marks)

(ii) The depth of the tissue interface is computed using a device which takes the whole echo time T as being spent on propagating to and fro in soft tissue only (i.e. calibrated at a speed of 1540 m s^{-1}). Determine by how much this **computed depth** differs from the **actual depth** of the tissue interface. (1 mark)

See p.114 for the Exam Report.

23 HKDSE 2019 Paper 2 Q.4(b)

2.2

The table shows information relating to the transmission of sound through different types of body tissues.

Tissue	Speed of sound / m s^{-1}	Acoustic impedance / kg m^{-2} s^{-1}
Bone	3780	7.15×10^6
Muscle	1590	1.65×10^6
Fat	1450	1.37×10^6

Table b

(a) Estimate the density of bone. (1 mark)

(b) When ultrasound is incident to a 'muscle-bone' boundary, find the ratio of the intensity of ultrasound reflected from the boundary to that incident to the boundary. (2 marks)

(c) Explain why in an ultrasound scan a 'muscle-bone' boundary is easier to be distinguished compared to a 'muscle-fat' boundary. (2 marks)

24 HKDSE 2021 Paper 2 Q.4(b)

2.3

An ultrasound transducer is used to scan an eye as shown in Figure u. The pulses reflected from interfaces A, B, C and D are recorded in the A-scan display below.

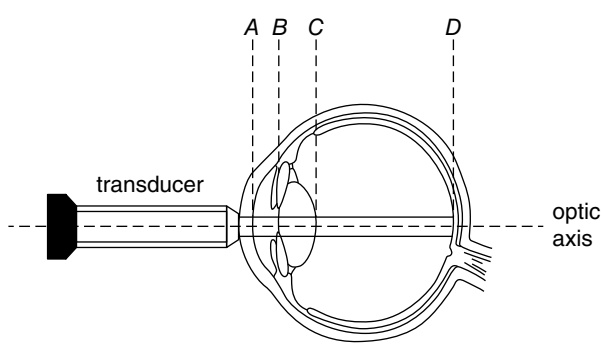


Fig u

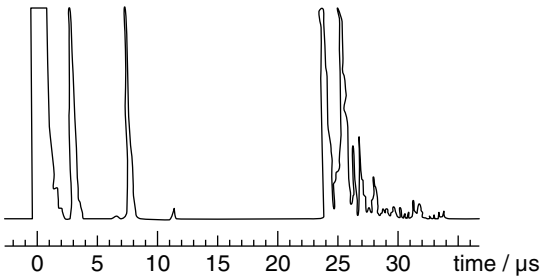


Fig v

(a) Estimate the thickness of the eye lens along the optic axis. Given: velocity of ultrasound in the eye lens = 1520 m s^{-1} (2 marks)

(b) Explain which frequency of ultrasound, 3 MHz or 15 MHz, is preferred for scanning the eye. (1 mark)

(c) Apart from forming images in a diagnostic scan, name **one** medical application of ultrasound. (1 mark)

25 HKDSE 2022 Paper 2 Q.4(a)

Misc

The table below shows some information of air, skin and muscle.

	Density (kg m^{-3})	Speed of sound (m s^{-1})
Air	1.20	340
Skin	1000	1520
Muscle	1040	1630

Table c

(a) Find the acoustic impedance of muscle. (1 mark)

(b) An ultrasound transducer is inclined at 5° to the normal of the skin surface when it is used for a scan. Find the angle of refraction of the ultrasound beam after it enters the skin from the air. (2 marks)

(c) Hence explain why an ultrasound transducer should be held perpendicular to the skin surface during a scan. (2 marks)

Q22 Exam Report: Candidates' performance in (a)(i) was fair. A few did not realise that the wavelength was twice the crystal's thickness. Many candidates failed to give concise explanations in (a)(ii). Few pointed out explicitly that the vibration of the crystal generates a 'voltage'. A lot of them just stated that an 'electrical signal' was produced when the ultrasound 'moves' the crystal, which was too general as an explanation. Many candidates did not understand what (b)(i) asked for, particularly the meaning of 'axial resolution'. Most candidates just answered how the frequency affected resolution and penetration or used generic terms such as 'clear' to describe the effects of a higher resolution. In (b)(ii), some candidates were not able to manipulate the calculation of inverse proportion. (c)(i) was well answered though a few candidates omitted the factor '2' for a round trip. Candidates' performance in (c)(ii) was fair as they did not realise that the true depth was 10 cm. Weaker candidates simply did not understand the relationship between the computed depth and the calibration speed.

Q24 Exam Report: In (a), candidates were asked to interpret an ultrasound A-scan display graph and extract relevant information from it to find the thickness of the eye lens. Most candidates were able to identify the time interval corresponding to the echoes produced at the boundaries of the eye lens and found the answer using the echo time formula. Some mistook the echo from the retina and obtained a centimetre thickness for the eye lens. This revealed that they had no idea about the size of the eye ball, which is about 2 cm in diameter. A few candidates made mistakes in manipulating microsecond (μs) time intervals. In (b), most candidates correctly chose 15 MHz rather than 3 MHz ultrasound for the A-scan as it provided a higher resolution. A few picked 3 MHz ultrasound and held that it was less harmful or it had a higher penetrating power. Less than half of the candidates were able to give examples of medical applications of ultrasound in (c) other than its diagnostic medical applications.

Q25 Exam Report: Most candidates did well in (a) although a few made mistakes in the unit of acoustic impedance. Many were able to answer (b) by using $\frac{\sin\theta_{air}}{\sin\theta_{skin}} = \frac{\text{speed of sound in air}}{\text{speed of sound in skin}}$. However, some wrongly substituted the speed(s) or employed the formula involving intensity reflection coefficients. (c) was poorly answered. Not many realised that the bending of the ultrasound beam implied from (b) would lead to distortions in the ultrasound image in terms of locations as well as brightness. Significant bending occurred even for a small deviation from the perpendicular position (the ultrasound speed in skin is almost five times that in air), thus echoes would no longer return to the transducer along the same straight path as one might have assumed.

Q23 Exam Report: Part (a) was well answered. In (b), a few misinterpreted the question and they calculated wrong ratios, e.g. the intensity of ultrasound transmitted through the boundary to that incident to the boundary or the ratio between the reflected and transmitted intensities. In (c), not many explicitly pointed out that a greater acoustic impedance difference between two tissues yields an easier distinguishable boundary due to a larger intensity reflection coefficient.

Know more

Hearing aids and electronic cochlear implants (p.72)

People with mild to moderate hearing loss can hear sounds better by wearing hearing aids (Fig a). A hearing aid is a small electronic device that fits in or behind the ear and amplifies sounds entering the ear so that the hair cells can detect them more easily.



Fig a A hearing aid.

A person who is deaf or has severe hearing loss may regain a sensation of hearing by having an electronic cochlear implant (Fig b). It consists of external parts that fit behind the ear and internal parts that are surgically placed under the skin. The sound is picked up by a microphone and then converted into electrical signals that are directly sent to the auditory nerves through an electrode array.

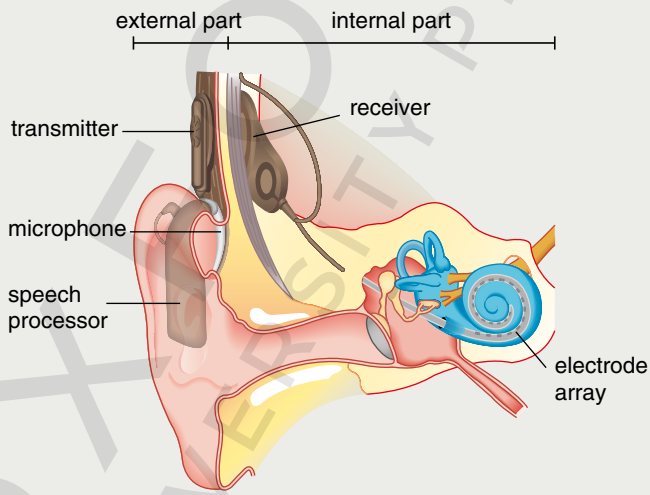


Fig b How an electronic cochlear implant is put into an ear.

Other medical uses of ultrasound (p.99)

Ultrasound enhances blood circulation around the fracture. This helps take away dead bone tissues and bring materials for building new bone tissues.

Apart from forming images of body organs, ultrasound can also be used for delivering energy. It has many other medical applications. Below are some examples.

- Break up kidney stones, gallstones etc.
- Break down cloudy lens in cataracted eyes
- Speed up bone healing after a fracture (Fig c)
- Remove tartar
- Eliminate tumours
- Heal wounds: relieve pain, promote circulation, diminish inflammation



Fig c